



BITS Pilani
Pilani Campus

Machine Learning

AIML CZG565

M3 : Linear Models for Regression

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BITS – CSIS - WILP

Disclaimer and Acknowledgement



- The content for these slides has been obtained from books and various other source on the Internet
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas
- We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation

Source: “Probabilistic Machine Learning, An Introduction”, Kevin P. Murphy, Slides of Prof.Sugata, Prof. Chetana from BITS Pilani, Prof. Raja vadhana from BITS Pilani , CS109 and CS229 stanford lecture notes and many others who made their course materials freely available online.

Course Plan

M1	Introduction
M2	Machine learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning
M7	Support Vector Machine
M8	Bayesian Learning
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

dataset

1

- Closed Form ✓

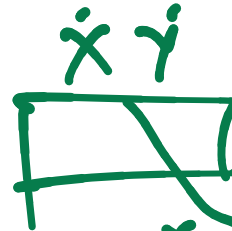


tion

bias

weight

Ne forgetti...



Learning Model Parameters

Gradient Descent Approach

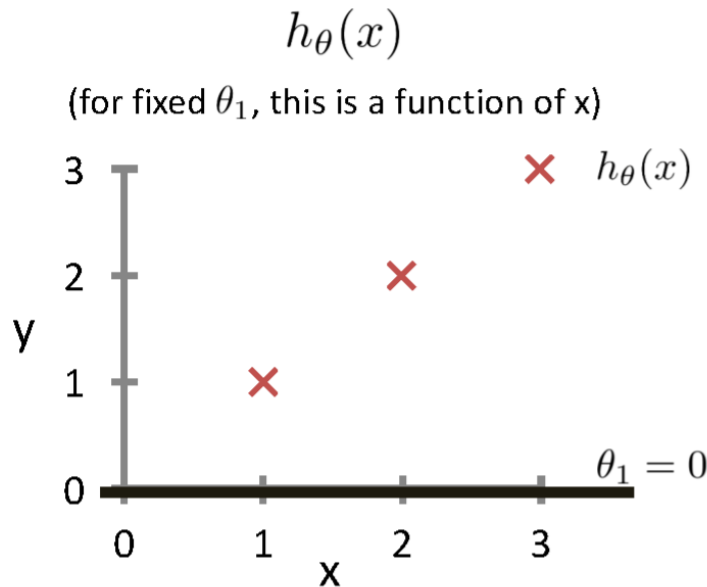
Intuition Behind Cost Function

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^n \left(h_{\theta} \left(x^{(i)} \right) - y^{(i)} \right)^2$$

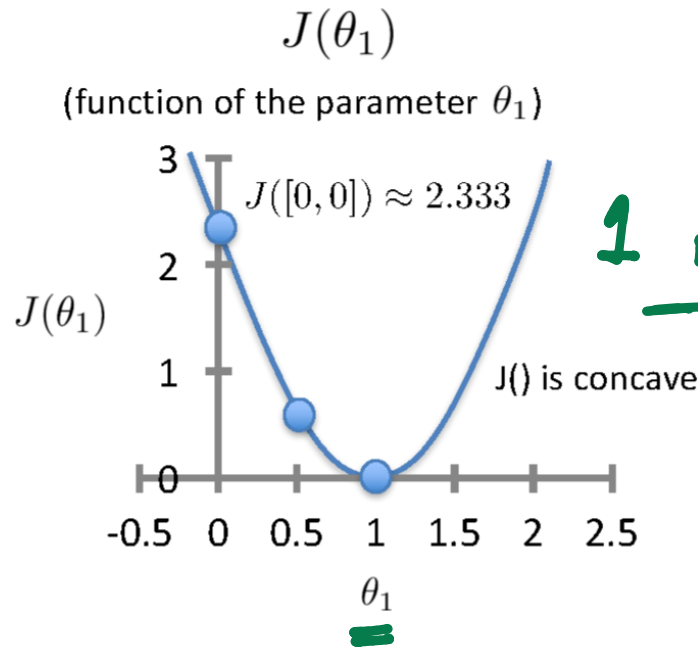
Residual

MSE

For insight on $J()$, let's assume $x \in \mathbb{R}$ so $\theta = [\theta_0, \theta_1]$

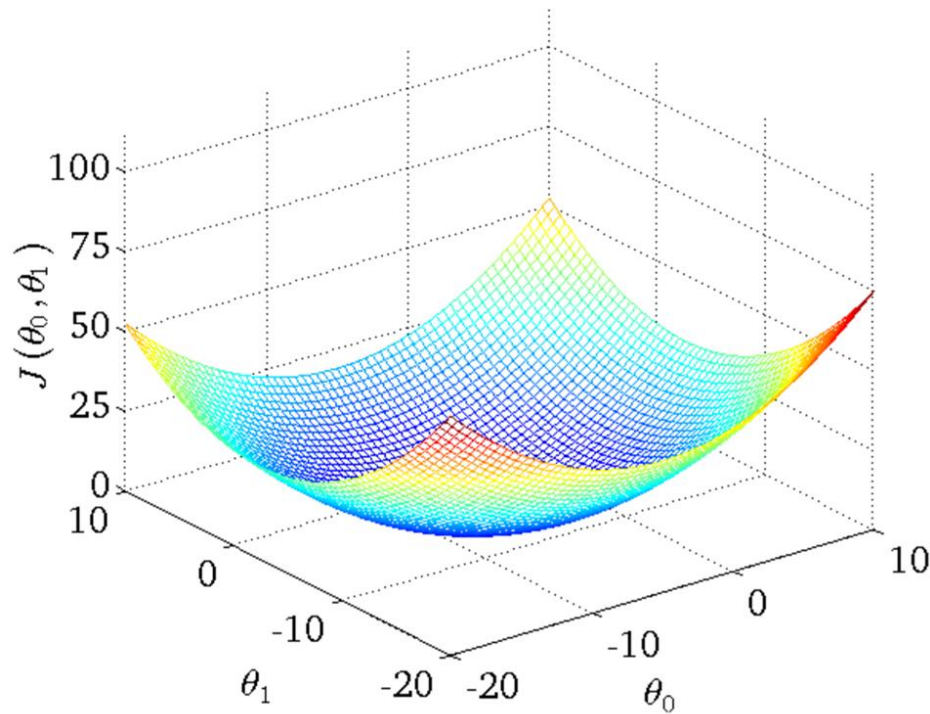


Source Credit : Based on example by Andrew Ng



1 minime

Intuition Behind Cost Function

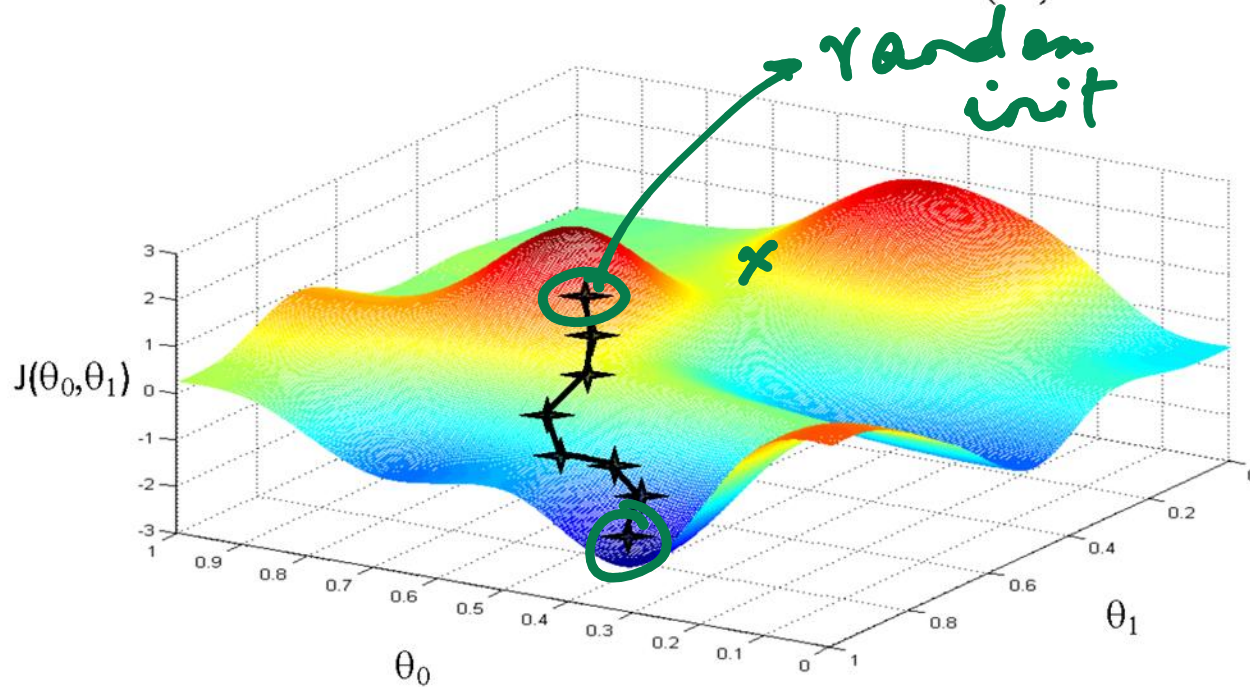


Source Credit : Based on example
by Andrew Ng

Basic Search Procedure

- Choose initial value for θ
- Until we reach a minimum:
 - Choose a new value for θ to reduce $J(\theta)$

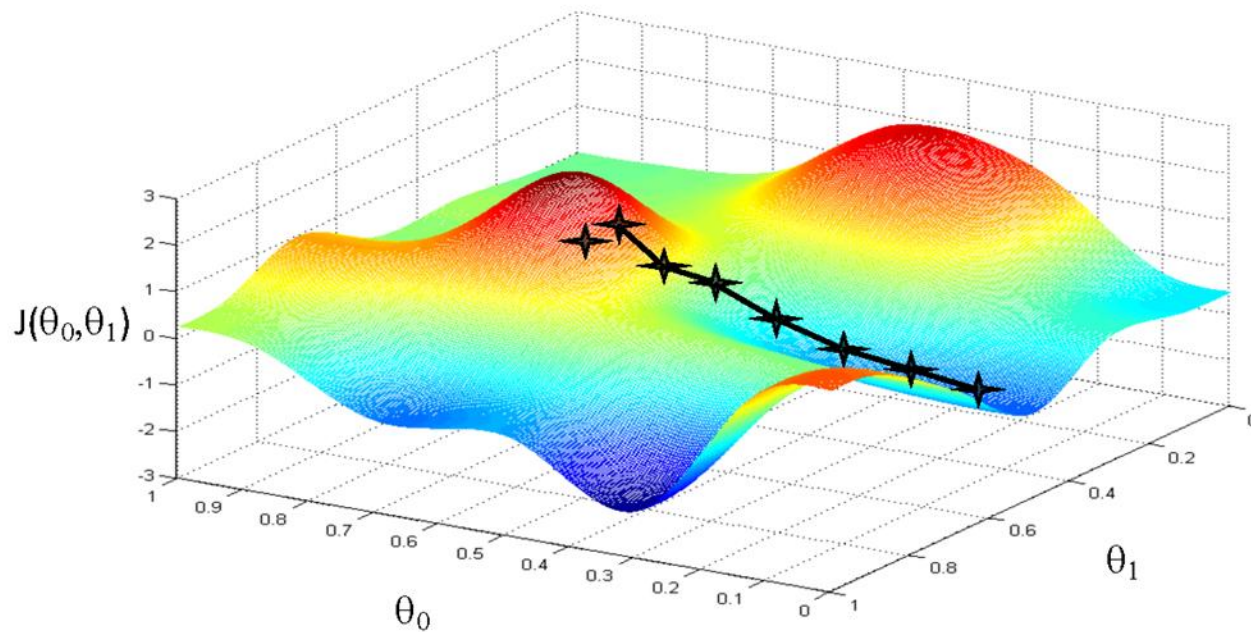
General
 $\min J(\theta)$



Source Credit : Figure by Andrew Ng

Basic Search Procedure

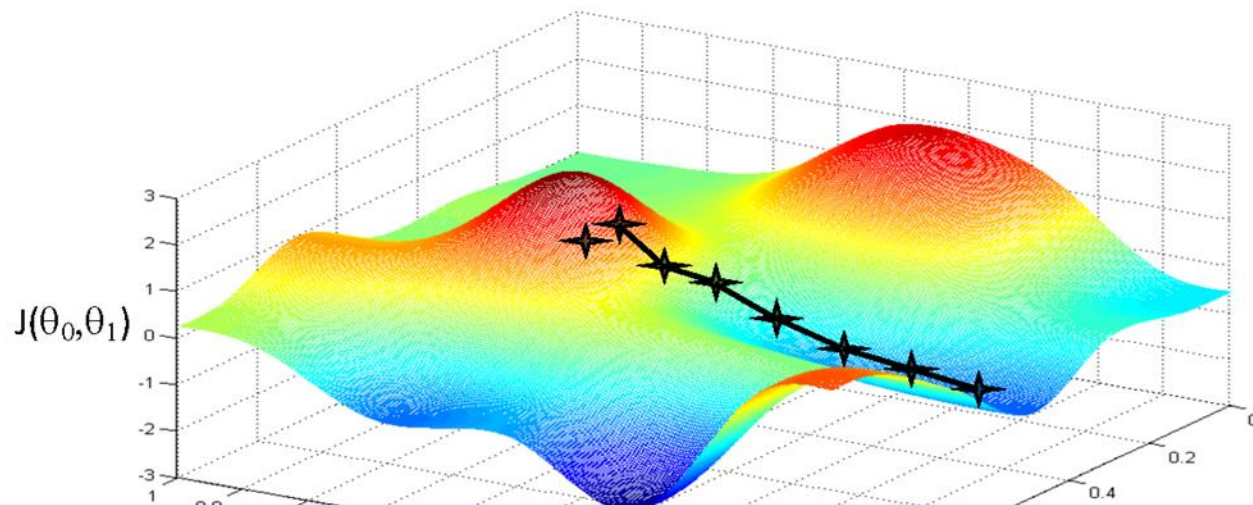
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Source Credit : Figure by Andrew Ng

Basic Search Procedure

- Choose initial value for θ
- Until we reach a minimum:
 - Choose a new value for θ to reduce $J(\theta)$



Since the least squares objective function is convex (concave),
we don't need to worry about local minima

Source Credit : Figure by Andrew Ng

Gradient Descent

Size in feet ² (x)	Price (\$) in 1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

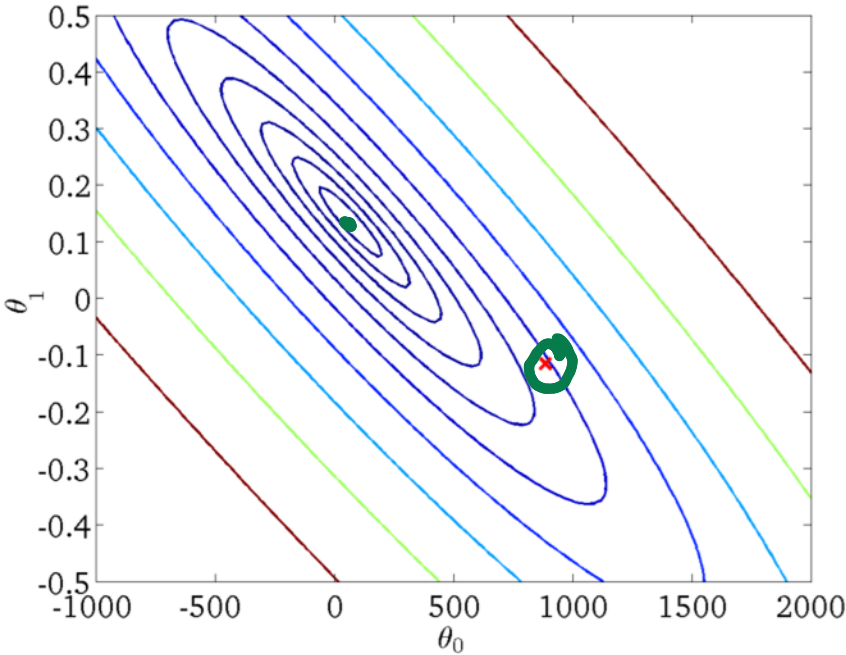
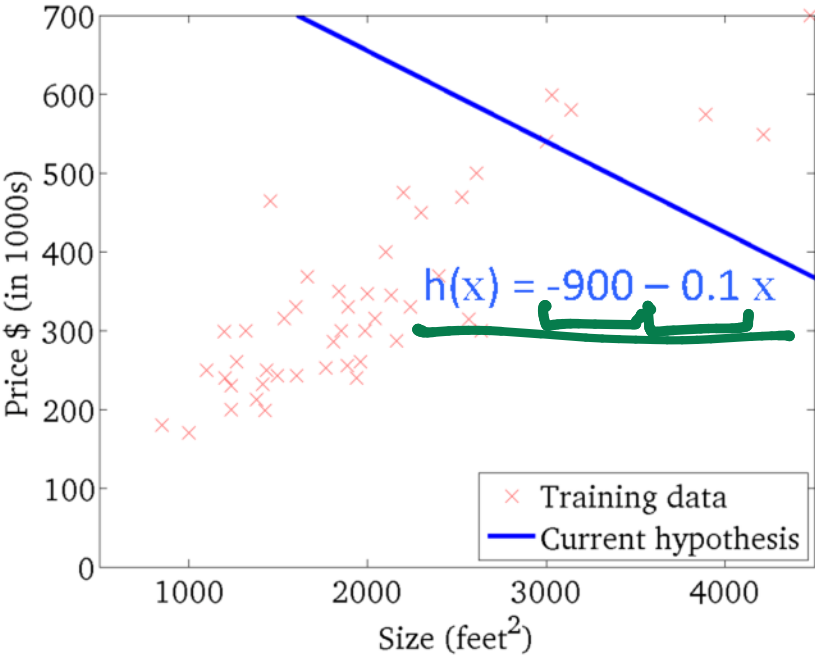
housing.

$h_{\theta}(x)$

$J(\theta_0, \theta_1)$

(for fixed θ_0, θ_1 , this is a function of x)

(function of the parameters θ_0, θ_1)

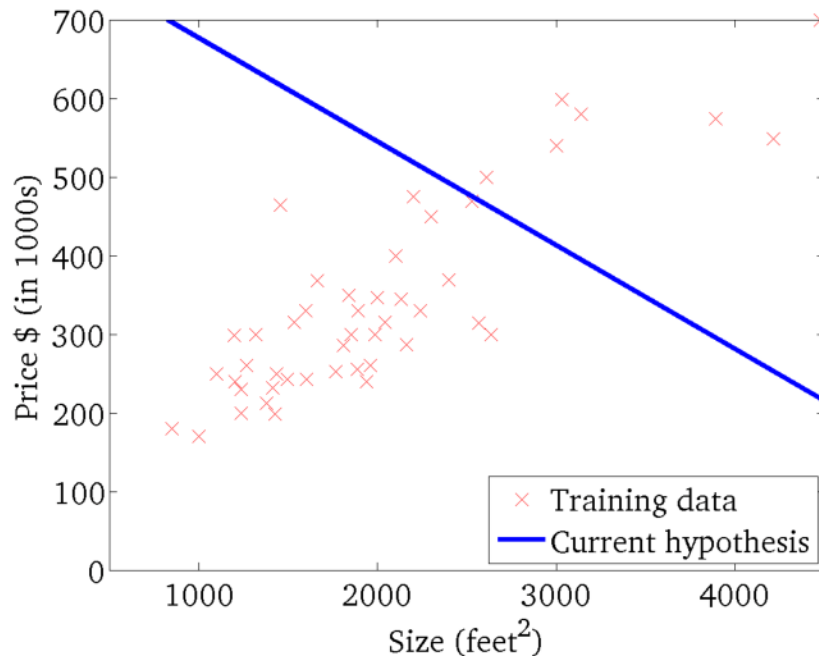


Source Credit : Slide by Andrew Ng

Gradient Descent

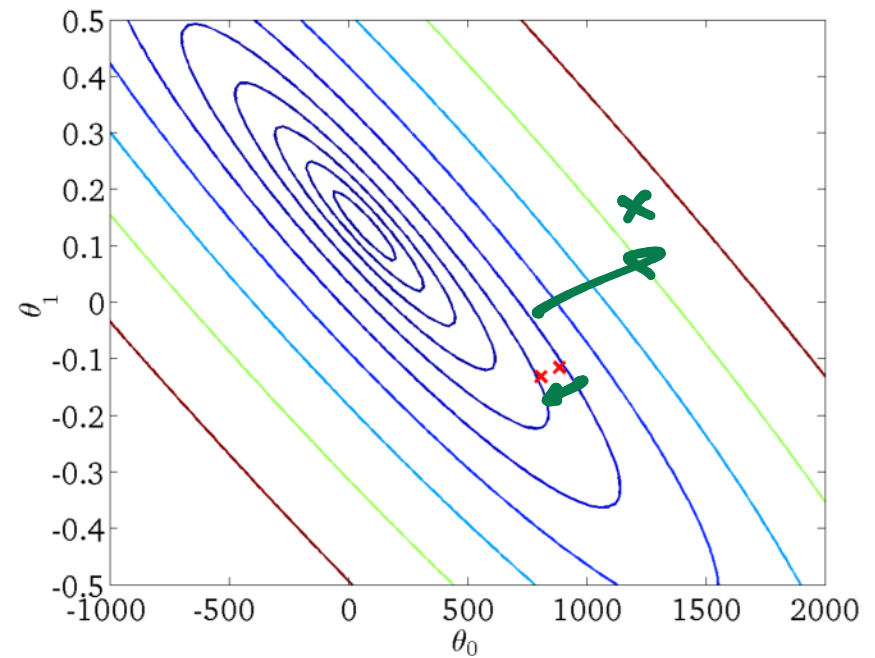
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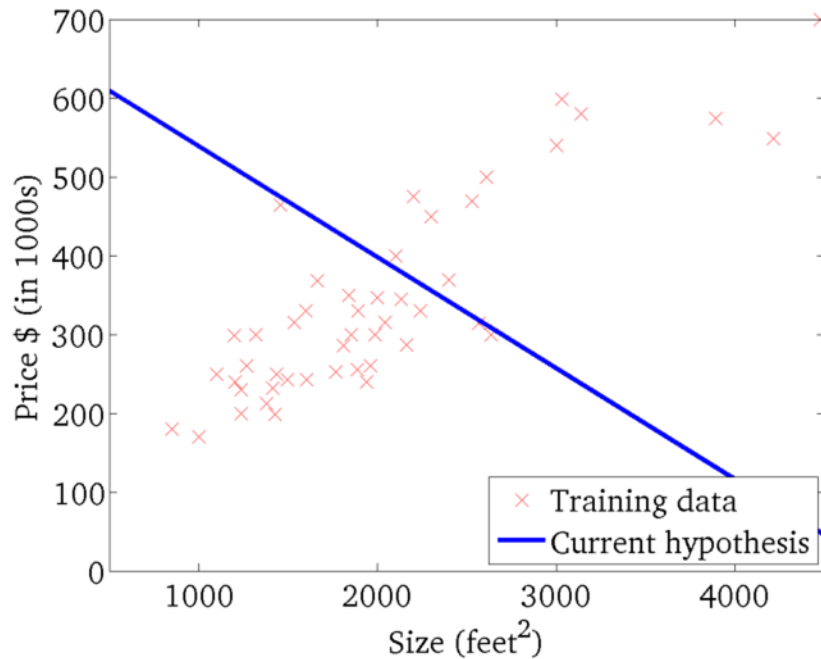


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Gradient Descent

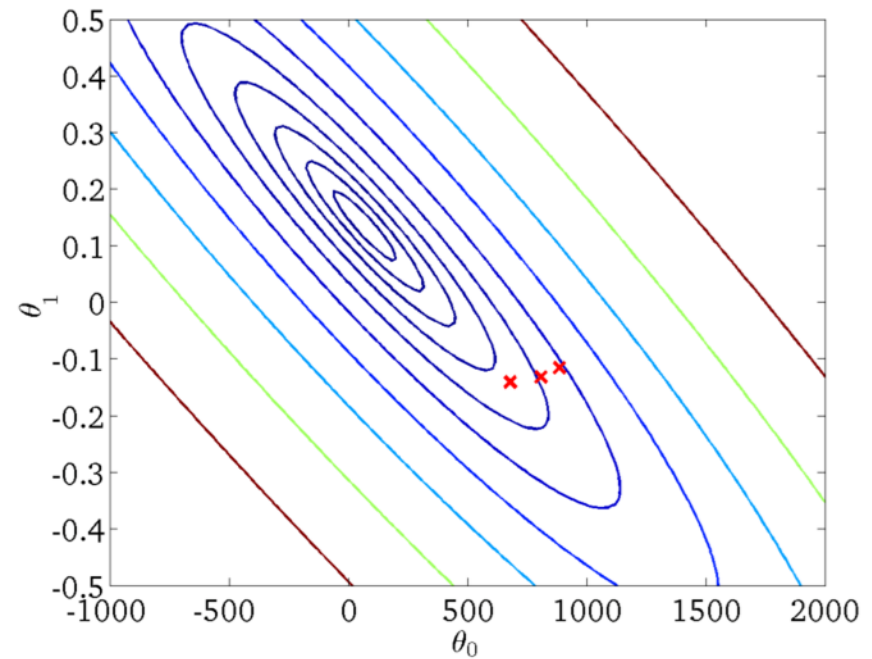
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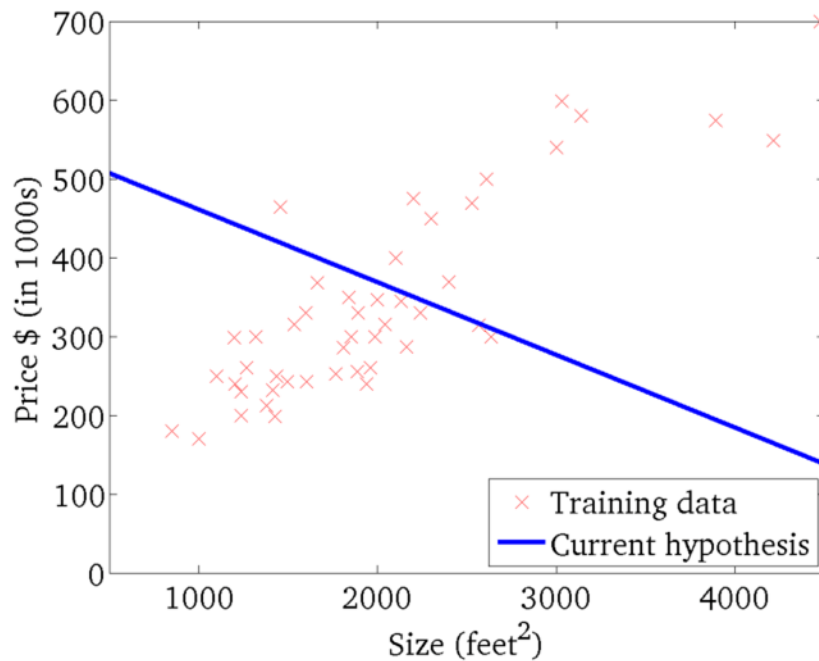


Source Credit : Slide by Andrew Ng

Gradient Descent

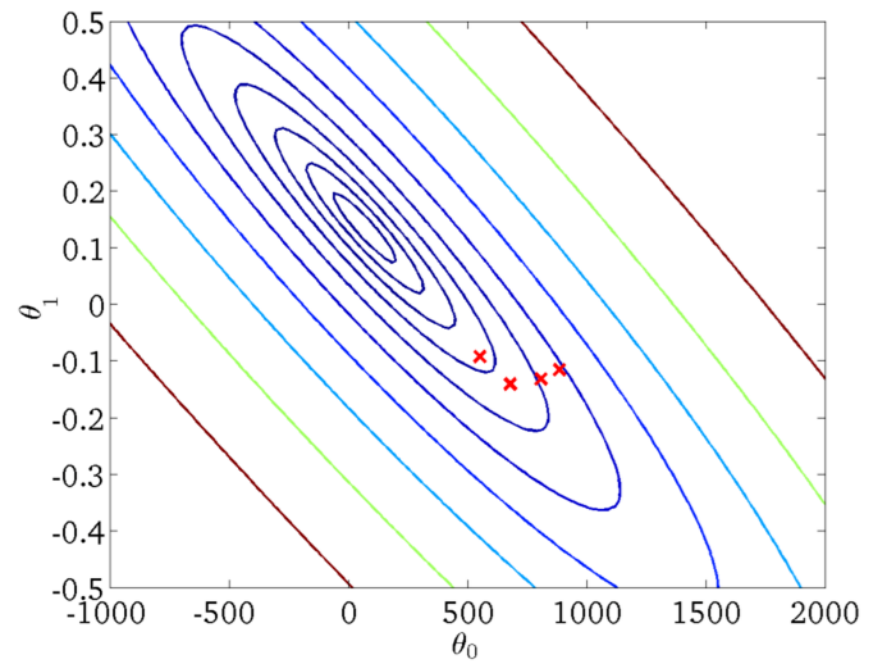
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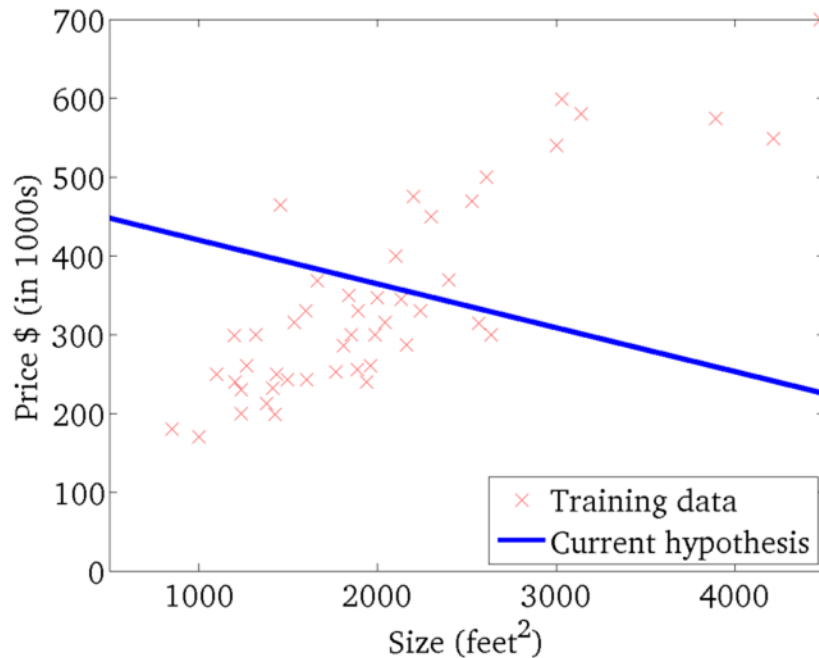


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Gradient Descent

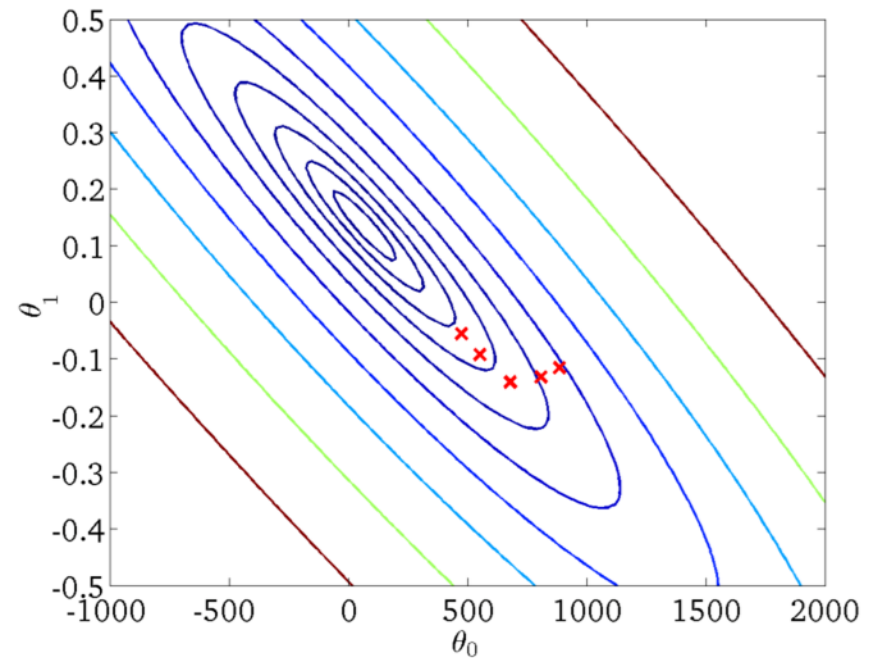
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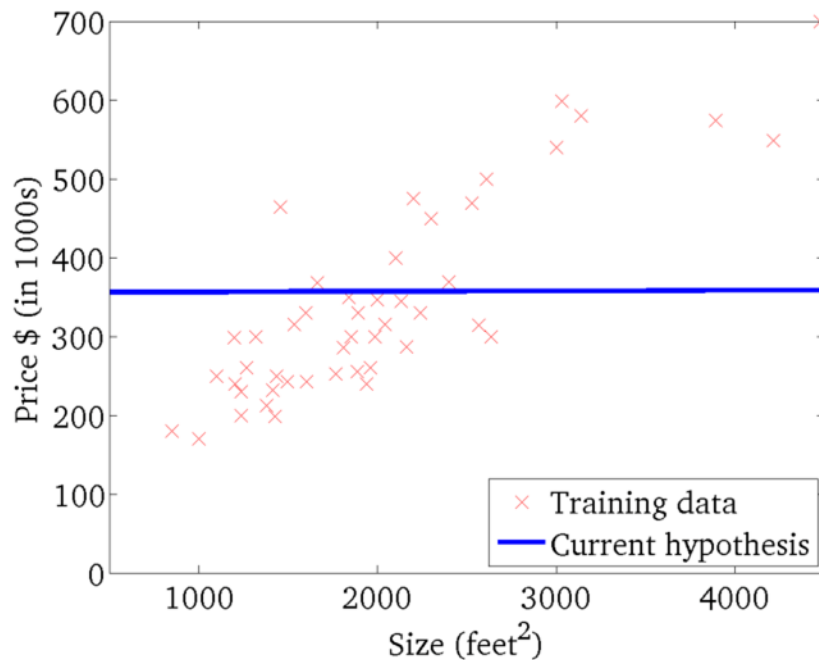


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Gradient Descent

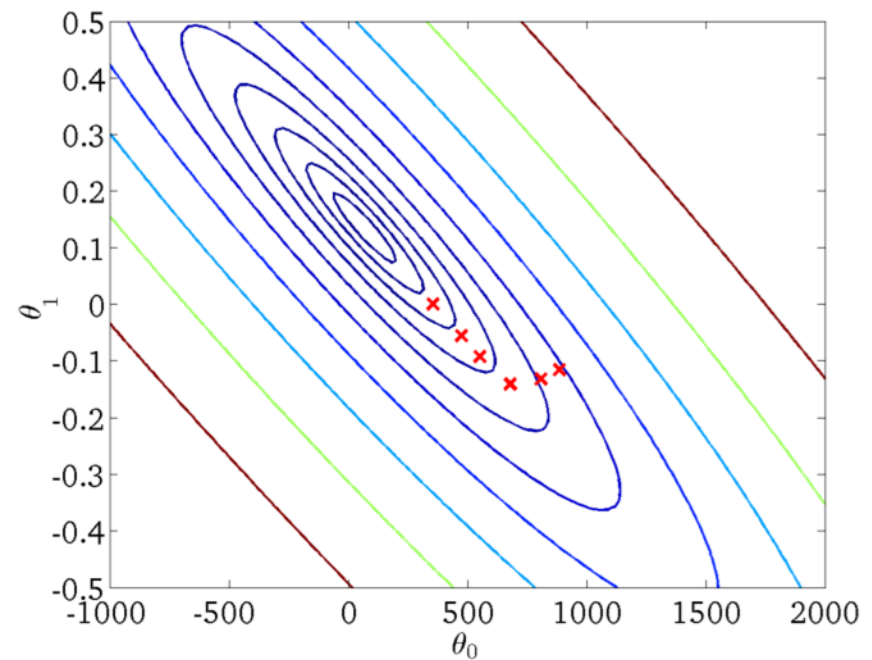
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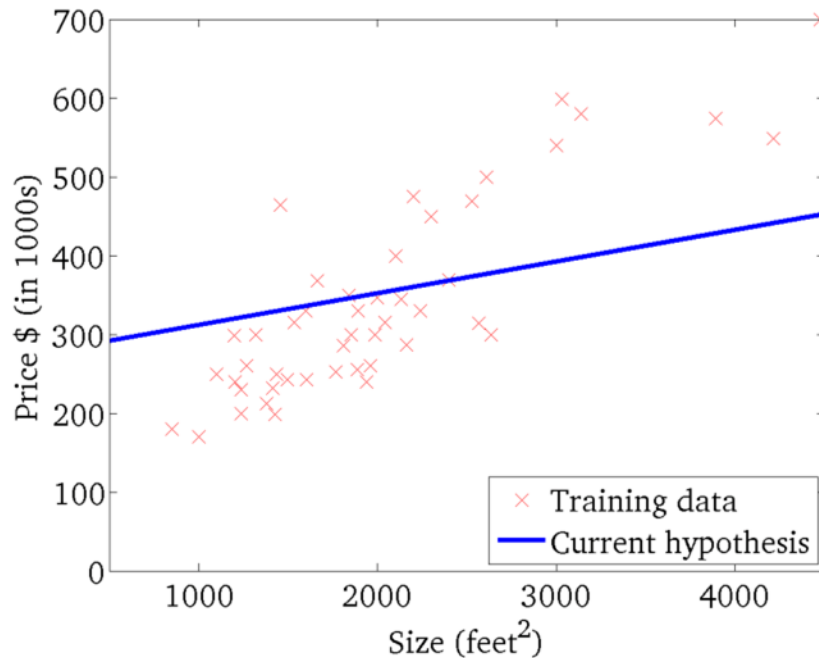


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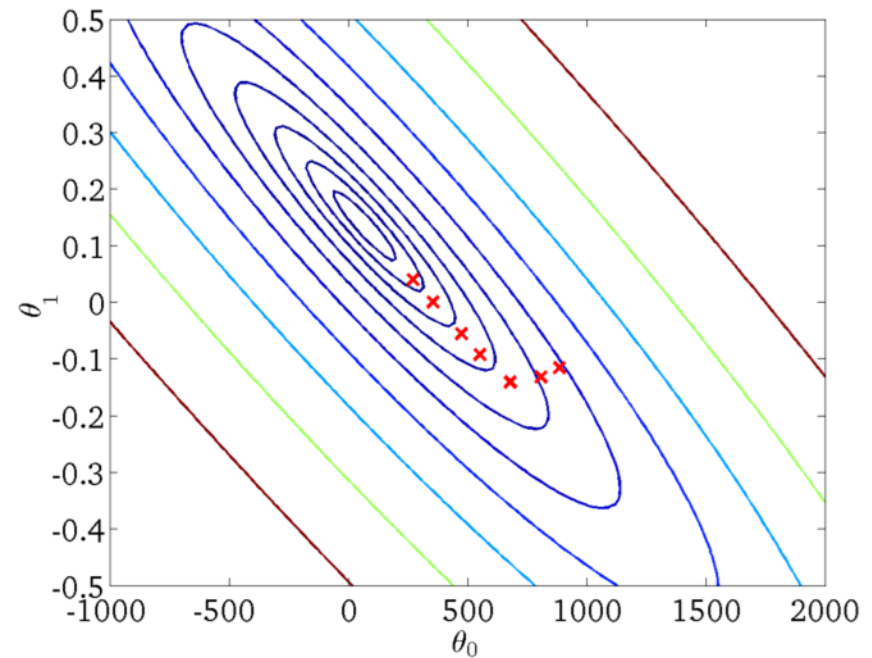
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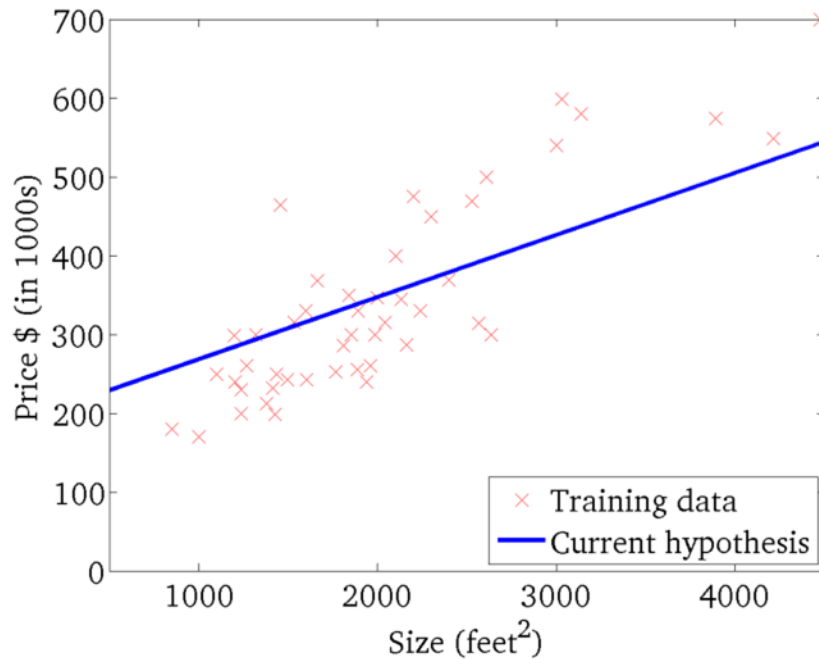


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Gradient Descent

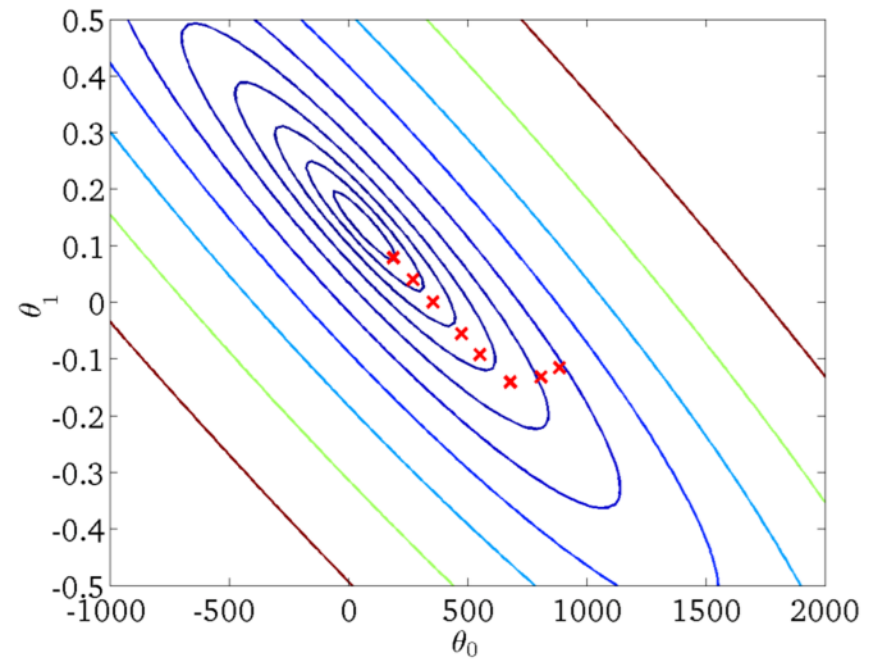
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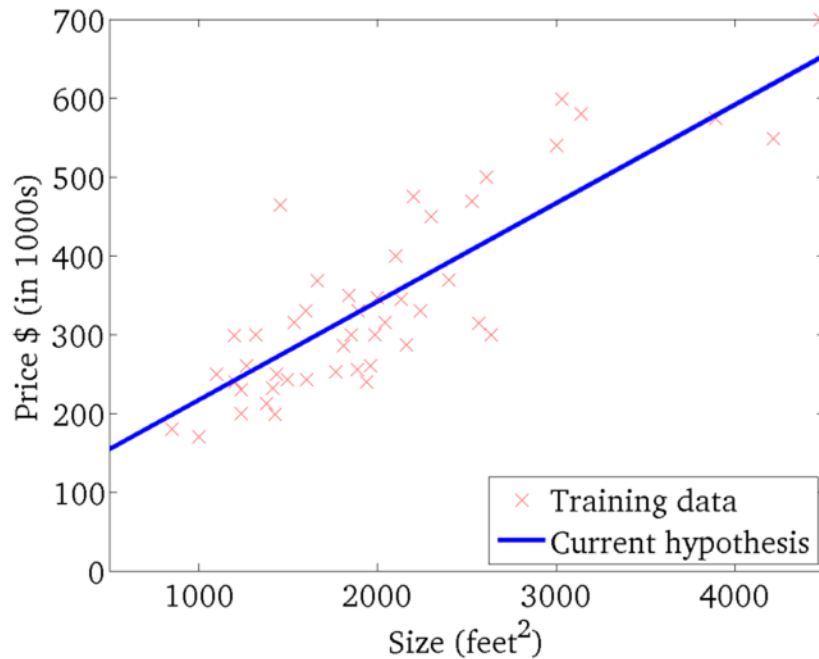


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Gradient Descent

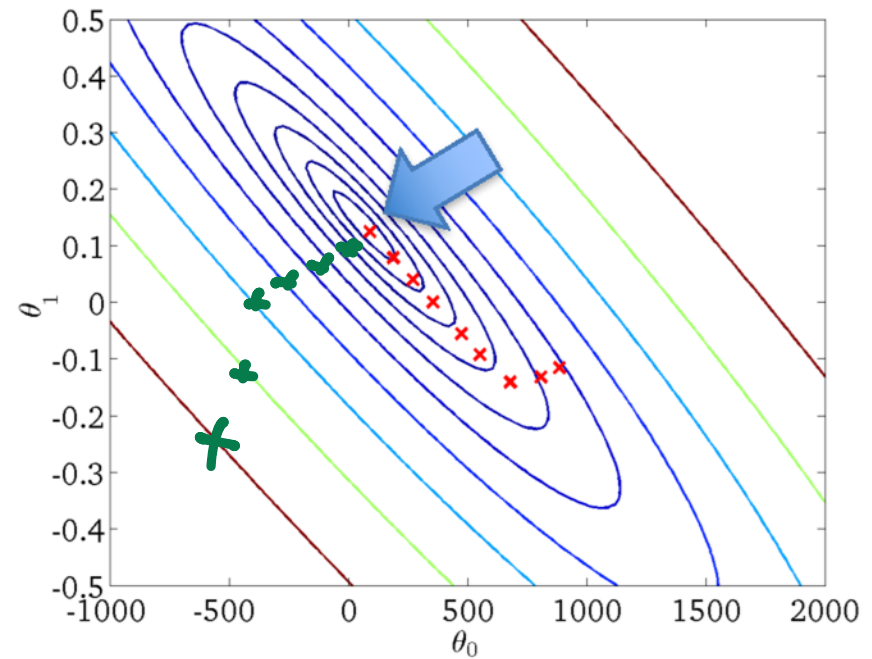
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(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



Source Credit : Slide by Andrew Ng



Gradient Descent

$$\theta_j = \theta_j - \alpha \cdot \frac{\partial J}{\partial \theta_j}$$

Handwritten notes: θ_j is labeled "Par" (Parameter) in red. α is labeled "Hy. Par" (Learning Rate) in red. The derivative term is labeled "innovate" in orange.

- Initialize θ
- Repeat until convergence

$$\theta_j \leftarrow \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

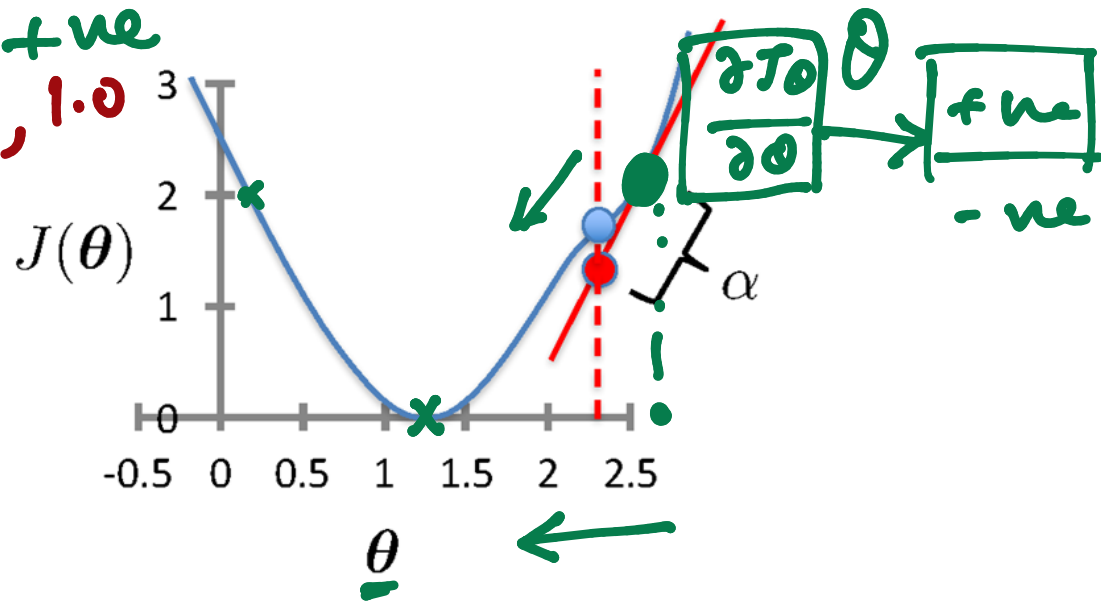
Handwritten notes: A red arrow points to the update rule. A green circle highlights the learning rate α .

simultaneous update
for $j = 0 \dots d$

step size $\leftarrow$ learning rate (small)
e.g., $\alpha = 0.05$

+ve slope $\rightarrow$ reduce α how much?

+ve, 0.1, 1.0



Gradient Descent

- Initialize θ
- Repeat until convergence

$$\theta_j \leftarrow \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

simultaneous update
for $j = 0 \dots d$

For Linear Regression:

$$\begin{aligned} \frac{\partial}{\partial \theta_j} J(\theta) &= \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(h_{\theta} \left(x^{(i)} \right) - y^{(i)} \right)^2 \\ &= \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(\sum_{k=0}^d \theta_k x_k^{(i)} - y^{(i)} \right)^2 \\ &= \frac{1}{n} \sum_{i=1}^n \left(\sum_{k=0}^d \theta_k x_k^{(i)} - y^{(i)} \right) \times \frac{\partial}{\partial \theta_j} \left(\sum_{k=0}^d \theta_k x_k^{(i)} - y^{(i)} \right) \\ &= \frac{1}{n} \sum_{i=1}^n \left(\sum_{k=0}^d \theta_k x_k^{(i)} - y^{(i)} \right) x_j^{(i)} \end{aligned}$$

Handwritten notes:

- Red bracket under the first term in the final equation: $\frac{1}{n} \sum_{i=1}^n \left(\hat{y}^{(i)} - y^{(i)} \right) x_j^{(i)}$
- Red bracket under the second term in the final equation: $\left(\sum_{k=0}^d \theta_k x_k^{(i)} - y^{(i)} \right) x_j^{(i)}$
- Red arrow pointing from the second term to the word "Num" (Numerator).
- Red bracket under the entire final equation.
- Red bracket under the entire derivation.

$d \rightarrow d+1 (Q)$



Gradient Descent for Linear Regression

θ α ϵ

- Initialize θ ✓
- Repeat until convergence ✓

$$\theta_j \leftarrow \theta_j - \alpha \frac{1}{n} \sum_{i=1}^n \left(h_{\theta} \left(x^{(i)} \right) - y^{(i)} \right) x_j^{(i)}$$

simultaneous update for $j = 0 \dots d$

- To achieve simultaneous update
 - At the start of each GD iteration, compute $h_{\theta} \left(x^{(i)} \right)$
 - Use this stored value in the update step loop
- Assume convergence when $\| \theta_{new} - \theta_{old} \|_2 < \epsilon$

Euclidean dist

L₂ norm: $\|v\|_2 = \sqrt{\sum_i v_i^2} = \sqrt{v_1^2 + v_2^2 + \dots + v_{|v|}^2}$

" < ϵ



together

θ_0 $\rightarrow$ update $\checkmark$

$\theta_1 \rightarrow \theta_1 - \alpha \frac{1}{n} \sum \underline{(\underline{h_0(x)} - y)x}$

.

:

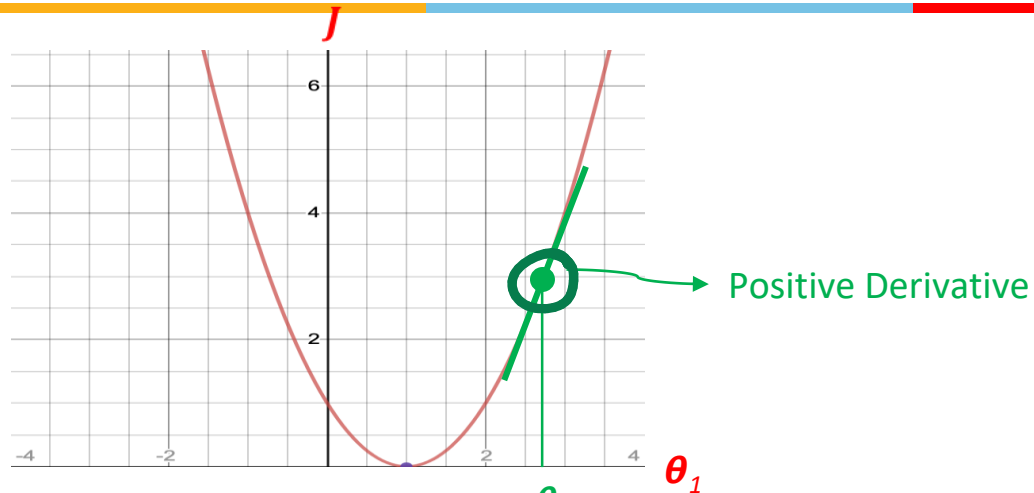
.

θ_d

$\theta_0 + \theta_1 x_1 + \dots$

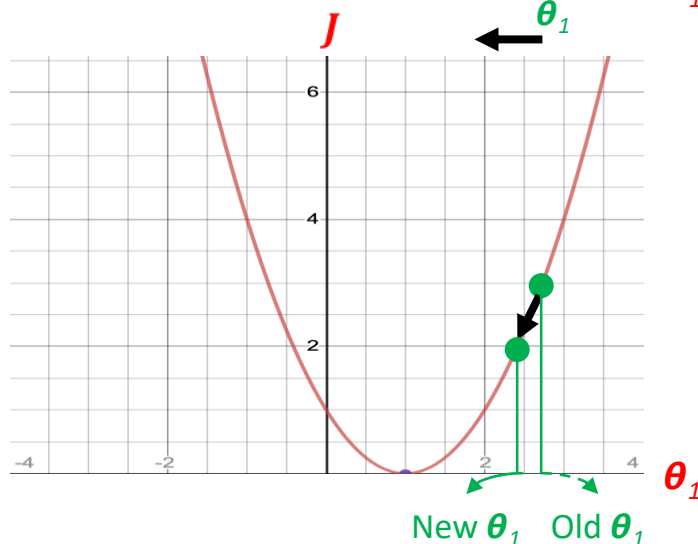
$h_0(x)$

Guarantee of Convergence



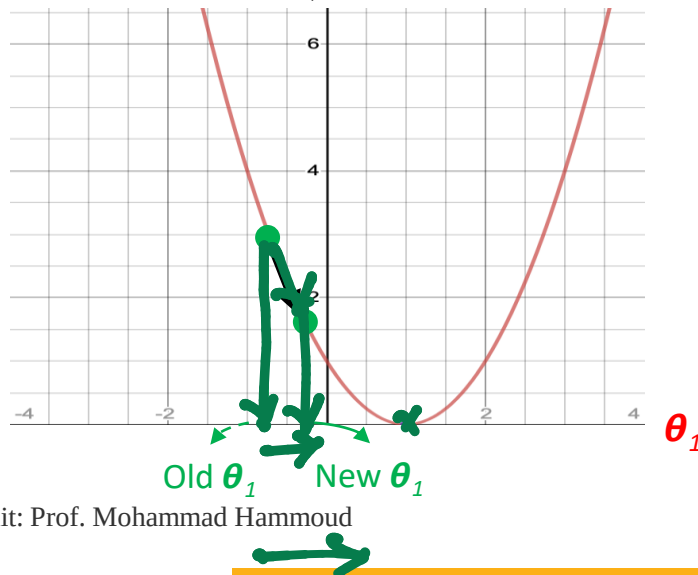
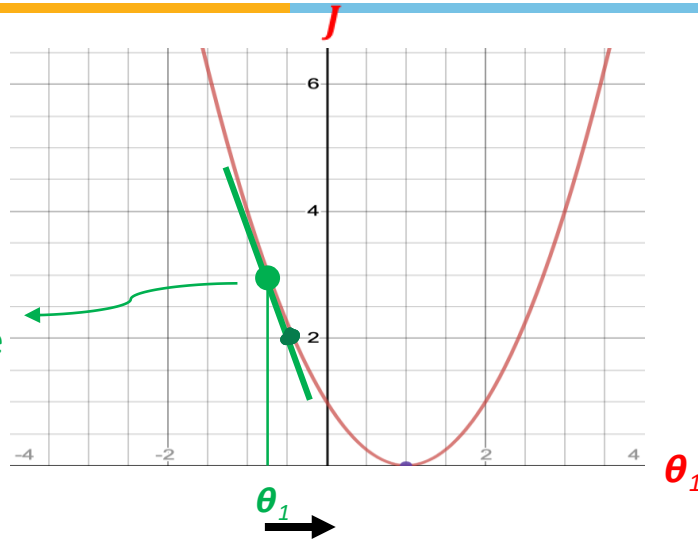
$$\begin{aligned}\theta_1 &= \theta_1 - \alpha \frac{dJ(\theta_1)}{d\theta_j} \\ &= \theta_1 - \alpha (\text{Positive Number})\end{aligned}$$

Decrease θ_1 by a certain value



Source Credit: Prof. Mohammad Hammoud

Guarantee of Convergence



$$\theta_1 = \theta_1 - \alpha \frac{dJ(\theta_1)}{d\theta_1}$$
$$= \theta_1 - \alpha (\text{Negative Number})$$

Increase θ_1 by a certain value

Left $\rightarrow$ +ve
-ve

$\theta_j \leftarrow \theta_j - \alpha (-ve)$
+ve inc in θ

Source Credit: Prof. Mohammad Hammoud

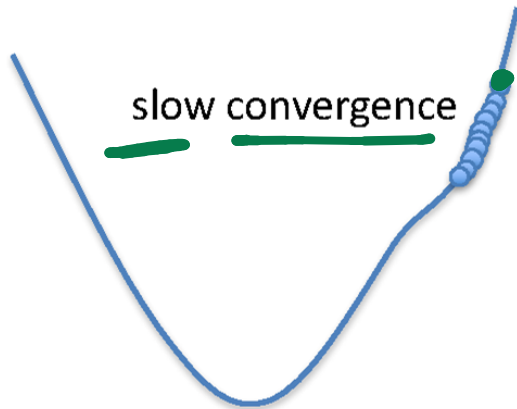
Choosing Learning Rate

→ Hy.P.

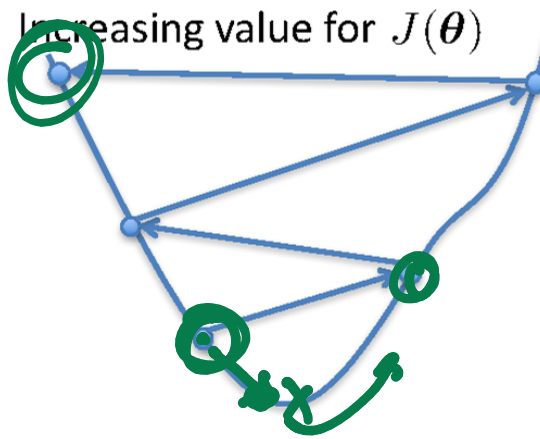
step size

α too small

slow convergence



α too large



Increasing value for $J(\theta)$

0.1
0.05
0.0001
1.0

(0.1, 0.05, 0.2)

- May overshoot the minimum
- May fail to converge
- May even diverge

To see if gradient descent is working, print out $J(\theta)$ each iteration

- The value should decrease at each iteration
- If it doesn't, adjust α

Gradient Descent algorithm : Effect of Feature Scaling

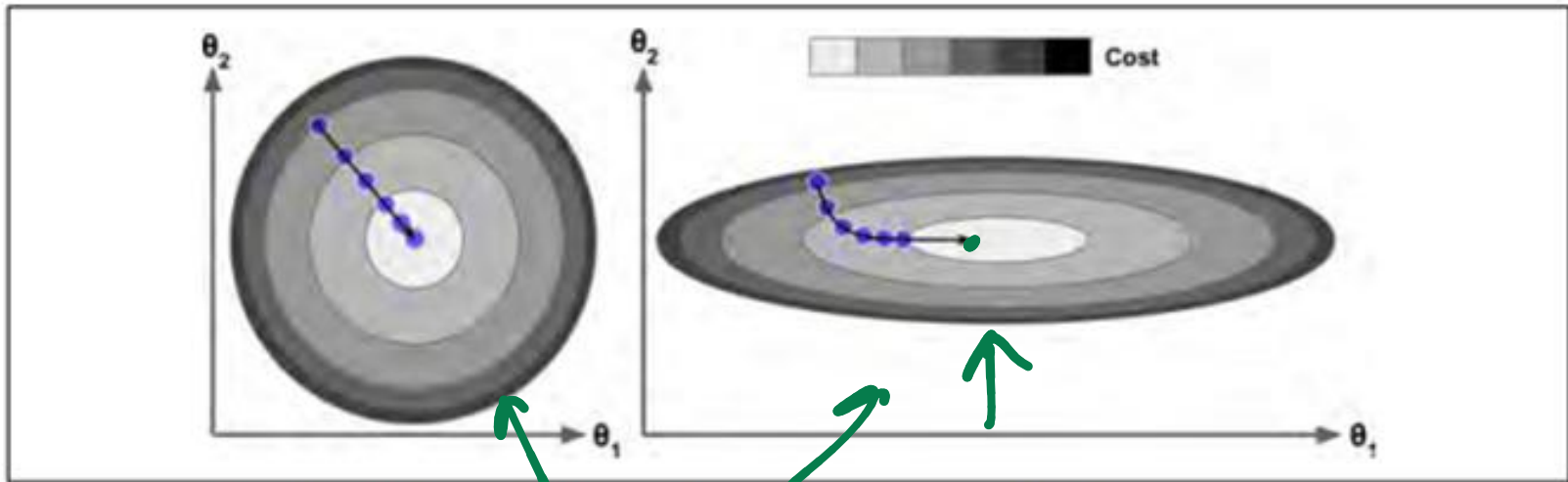


Figure 4-7. Gradient Descent with and without feature scaling

$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 \cdot \dots + \theta_d x_d$$

0-10
1000
1000

Gradient Descent algorithm : Effect of Learning Rate :



Problem Type 2 – Interpretation of Convergence or Effect of Hyperparameters

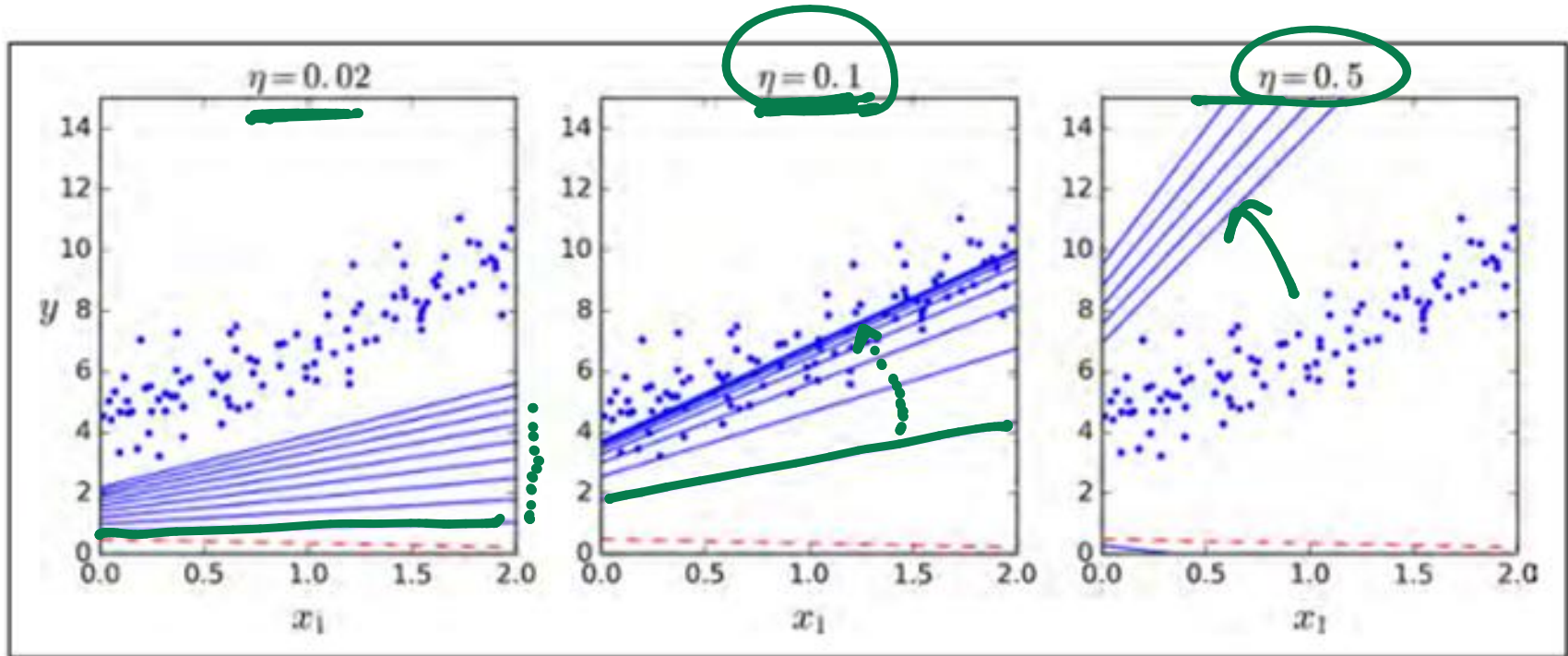


Figure 4-8. Gradient Descent with various learning rates

$\alpha, \eta \rightarrow \underline{\underline{LR}}$



Fit a linear Regression Line :

Gradient Descent : Problem Type 3

w, x, y

Steps :
(Assuming : 'n' no.of.instances and two predictors {x1, x2} and linear regression)

1. Identification of the equations $y = w_0 + w_1x_1 + w_2x_2$

2. Cost function & derivative

init $0, w$

$h_0(x)$

actual y

init $0, w$

$h_0(x)$

Bias

1. $w_0' = w_0 - \frac{1}{n} * \text{learning rate} * (\text{sum}(w_0 + w_1x_1 + w_2x_2 - y))$ (1)

2. $w_1' = w_1 - \frac{1}{n} * \text{learning rate} * (\text{sum}(w_0 + w_1x_1 + w_2x_2 - y) * x_1)$

3. $w_2' = w_2 - \frac{1}{n} * \text{learning rate} * (\text{sum}(w_0 + w_1x_1 + w_2x_2 - y) * x_2)$

3. Apply the equations

update 0

$\frac{1}{n} \sum_{i=1}^d (\hat{y} - y) x$

all data points

$y \rightarrow \text{Actual}$

$h_0(x) = \hat{y}$

pred

Fit a linear Regression Line :

Gradient Descent

Fit a linear regression. Show only the first iteration of Gradient descent algorithm using learning rate of 0.02 for the following data , if the Relative Risk of Coronary Heart Disease is believed to be only linearly dependent on BMI as well as Diastolic Pressure. Assume the intercept of the regression model as 5 and the slope of independent variables as -0.03 (negative). → target

θ_0, w_0, b

Patient	Systolic Pressure mm Hg	Diastolic Pressure mm Hg	BMI	Waist Circumference Threshold cm	RR-CHD (Relative Risk of Coronary Heart Disease)
1	140	80	35	100	1.81
2	120	80	25	80	1.22
3	130	100	30	60	1.71

Steps :

1. Identification of the equations $y = w_0 + w_1X_1 + W_2X_2$
2. Cost function & derivative
 1. $W_0' = w_0 - 1/3 * \text{learning rate} * (\text{sum}(w_0 + w_1X_1 + W_2X_2 - y))$
 2. $W_1' = w_1 - 1/3 * \text{learning rate} * (\text{sum}(w_0 + w_1X_1 + W_2X_2 - y) * x_1)$
 3. $W_2' = w_2 - 1/3 * \text{learning rate} * (\text{sum}(w_0 + w_1X_1 + W_2X_2 - y) * x_2)$
3. Apply the equations

$\alpha = 0.02$

$\theta_0 = 5$

$\theta_1 = -0.03$

$\theta_2 = -0.03$

Fit a linear Regression Line :

Gradient Descent

$$y = mx + c$$

Fit a linear regression. Show only the first iteration of Gradient descent algorithm using learning rate of **0.02** for the following data , if the Relative Risk of Coronary Heart Disease is believed to be only linearly dependent on BMI as well as Diastolic Pressure. Assume the intercept of the regression model as **5** and the slope of independent variables as **-0.03 (negative)**.

Patient	Systolic Pressure mm Hg	Diastolic Pressure mm Hg	BMI	Waist Circumference Threshold cm	RR-CHD (Relative Risk of Coronary Heart Disease)
1	140	80	35	100	1.81
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3	130	100	30	60	1.71

Steps :

1. Identification of the equations:

$$h_0(x) = w_0 + w_1 x_1 + w_2 x_2$$

$$RRCHD = 5 - 0.03 BMI - 0.03 DP$$

0₁

Fit a linear Regression Line : Gradient Descent

Fit a linear regression. Show only the first iteration of Gradient descent algorithm using learning rate of **0.02** for the following data , if the Relative Risk of Coronary Heart Disease is believed to be only linearly dependent on BMI as well as Diastolic Pressure. Assume the intercept of the regression model as **5** and the slope of independent variables as **-0.03 (negative)**.

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1	140	80	35	100	1.81
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Steps :

1. Identification of the equations:

GENERAL FORM

$$y = w_0 + w_1 \cdot x_1 + w_2 \cdot x_2$$

FOR THIS PROBLEM

$$\text{RR-CHD} = w_0 + w_1 \cdot \text{BMI} + w_2 \cdot \text{DiastolicPressure}$$

SUBSTITUTING INITIAL WEIGHTS

$$\hat{y} = 5 + (-0.03) \cdot \text{BMI} + (-0.03) \cdot \text{DiastolicPressure}$$

Fit a linear Regression Line : Gradient Descent



Step 2: Weight Update

$$W = W - \alpha \frac{1}{n} \sum (y - \hat{y})^2$$

2A. Compute predictions ($\hat{y}$) and errors for each patient

FORMULA USED

$$\hat{y} = 5 - 0.03 \cdot x_1 - 0.03 \cdot x_2$$

$$\text{error} = \hat{y} - y$$

Pati ent	x1 (BMI)	x2 (Diastolic)	y (actual)	0.03·x1	0.03·x2	$\hat{y} = 5 - 0.03 \cdot x_1 - 0.03 \cdot x_2$	<u>error = $\hat{y} - y$</u>
1	35	80	1.81	1.05	2.40	$5 - 1.05 - 2.40 = 1.55$	$1.55 - 1.81 = -0.26$ x35
2	25	80	1.22	0.75	2.40	$5 - 0.75 - 2.40 = 1.85$	$1.85 - 1.22 = +0.63$ x25
3	30	100	1.71	0.90	3.00	$5 - 0.90 - 3.00 = 1.10$	$1.10 - 1.71 = -0.61$ x30

$\sum$

Fit a linear Regression Line : Gradient Descent

2B. Compute the three required sums

Each weight update uses a different weighted sum of the prediction errors.

$$\begin{aligned} h_0(x) &= \hat{y} \\ \hat{y} - y \\ \sum (\hat{y} - y) x \end{aligned}$$

$$\sum (\hat{y} - y) x_0$$

SUM A — used to update W_0

$$\begin{aligned} &\sum (\hat{y} - y) \cdot 1 \\ &= (-0.26) + (0.63) + (-0.61) \\ &= -0.24 \end{aligned}$$

$$\text{Sum A} = -0.24$$

Σ

SUM B — used to update W_1
(BMI)

$$\begin{aligned} &\sum (\hat{y} - y) \cdot x_1 \\ &= (-0.26) \times 35 + (0.63) \times 25 \\ &\quad + (-0.61) \times 30 \\ &= -9.10 + 15.75 + (-18.30) \\ &= -11.65 \end{aligned}$$

$$\text{Sum B} = -11.65$$

SUM C — used to update W_2
(Diastolic)

$$\begin{aligned} &\sum (\hat{y} - y) \cdot x_2 \\ &= (-0.26) \times 80 + (0.63) \times 80 + (-0.61) \times 100 \\ &= -20.80 + 50.40 + (-61.00) \\ &= -31.40 \end{aligned}$$

$$\text{Sum C} = -31.40$$

Fit a linear Regression Line : Gradient Descent

Step 3. Apply the weight update rule and Solve the equations

$$w = \alpha \cdot \frac{1}{n} \sum$$

GENERAL UPDATE RULE

$$w' = w - (\alpha / n) \times \text{Sum}$$

$$\alpha / n = 0.02 / 3 = 0.006667$$

Updating W0

uses Sum A = -0.24

$$W0' = 5 - 0.006667 \times (-0.24)$$

$$= 5 - (-0.0016)$$

$$= 5 + 0.0016$$

$$W0' = 5.0016$$

Updating W1 (BMI)

uses Sum B = -11.65

$$W1' = -0.03 - 0.006667 \times (-11.65)$$

$$= -0.03 - (-0.0777)$$

$$= -0.03 + 0.0777$$

$$W1' = 0.0477$$

Updating W2 (Diastolic)

uses Sum C = -31.40

$$W2' = -0.03 - 0.006667 \times (-31.40)$$

$$= -0.03 - (-0.2093)$$

$$= -0.03 + 0.2093$$

$$W2' = 0.1793$$

Fit a linear Regression Line : Gradient Descent

UPDATED MODEL EQUATION

1st iter.

$$\text{RR-CHD} = 5.0016 + 0.0477 \text{ BMI} + 0.1793 \text{ DiastolicPressure}$$

KEY OBSERVATIONS

1. W_0 barely changed ($5 \rightarrow 5.0016$) — the intercept was already close to the correct value.
2. W_1 and W_2 both flipped from -0.03 to positive values, showing the model was over-penalising both BMI and Diastolic Pressure.
3. Further iterations of gradient descent would continue to reduce the cost function until convergence.

init $\rightarrow$ par + hy. par

$$h_{\theta}(x)$$

$$\hat{y}$$

$$\hat{y} - y$$

$$(\hat{y} - y)x$$

$$w_0 \rightarrow 1$$

$$w_1 \rightarrow x_1$$

$$w_2 \rightarrow x_2$$

1 iter

$$\sum$$

$$\downarrow$$

$$w' = w - \frac{\lambda}{n} \sum$$

$$\frac{\text{init}}{\text{①}} J(\theta) \rightarrow J(0)$$

Closed Form Solution Vs. Gradient Descent

Gradient Descent	Closed Form Solution
• Requires multiple iterations • Need to choose α • Works well when n is large • Can support incremental learning	• Non-iterative • No need for α • Slow if n is large – Computing $(X^T X)^{-1}$ is roughly $O(n^3)$

Types of Gradient Descent Algorithms

Gradient Descent: Variants

- **Batch** gradient descent refers to calculating the derivative from all training data before calculating an update.
- **Minibatch** refers to calculating derivative of mini groups of training data before calculating an update.
- **Stochastic** gradient descent refers to calculating the derivative from each training data instance and calculating the update immediately

repeat until convergence {

$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

}

$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$

Update θ_0 and θ_1 simultaneously

$$\text{temp0} := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

$$\text{temp1} := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x^{(i)}$$

$$\theta_0 := \text{temp0}$$

$$\theta_1 := \text{temp1}$$

$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$

Gradient Descent: Variants

- **Batch** gradient descent refers to calculating the derivative from all training data before calculating an update.

```

Initialize the Parameters  $(\theta_0^1, \theta_1^1, \dots)$ 
K=1
Repeat until Convergence {
     $\theta_0^{k+1} = \theta_0^k - \frac{\alpha}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$ 
     $\theta_1^{k+1} = \theta_1^k - \frac{\alpha}{m} \sum_{i=1}^m ((h_{\theta}(x^{(i)}) - y^{(i)}) * x_1^{(i)})$ 
     $\theta_2^{k+1} = \theta_2^k - \frac{\alpha}{m} \sum_{i=1}^m ((h_{\theta}(x^{(i)}) - y^{(i)}) * x_2^{(i)})$ 
    .....
K=k+1
}
return  $(\theta_0^1, \theta_1^1, \dots)$ 
    
```

Gradient Descent: Variants

- **Stochastic gradient** descent refers to calculating the derivative from each training data instance and calculating the update immediately

Randomly shuffle training instances

Initialize the Parameters $(\theta_0^1, \theta_1^1, \dots)$

$K=1$

Repeat until Convergence {

Sample with replacement, only one random training instance "i" at a time

$$\begin{aligned}\theta_0^{k+1} &= \theta_0^k - \alpha (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x \\ \theta_1^{k+1} &= \theta_1^k - \alpha (h_{\theta}(x^{(i)}) - y^{(i)}) * x_1^{(i)} \\ \theta_2^{k+1} &= \theta_2^k - \alpha (h_{\theta}(x^{(i)}) - y^{(i)}) * x_2^{(i)}\end{aligned}$$

.....

$K=k+1$

}

}

return $(\theta_0^1, \theta_1^1, \dots)$

$$\begin{aligned}x^{(i)} &\rightarrow \theta_s \rightarrow y^{(i)} \\ x_1^{(i)} & \\ x_2^{(i)} & \\ &\vdots\end{aligned}$$

Gradient Descent: Variants

- **Minibatch** refers to calculating derivative of mini groups of training data before calculating an update.

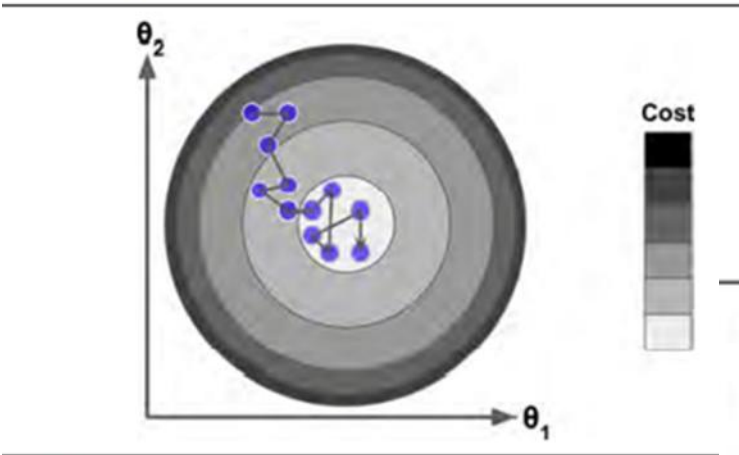
Divide the training instances into "N" batches each of size "m"
 Initialize the Parameters ($\theta_0^1, \theta_1^1, \dots$)
 K=1
 Repeat until Convergence {
 Repeat for every batch in 1 : N, each with 'm' instances {
 $\theta_0^{k+1} = \theta_0^k - \frac{\alpha}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$
 $\theta_1^{k+1} = \theta_1^k - \frac{\alpha}{m} \sum_{i=1}^m ((h_{\theta}(x^{(i)}) - y^{(i)}) * x_1^{(i)})$
 $\theta_2^{k+1} = \theta_2^k - \frac{\alpha}{m} \sum_{i=1}^m ((h_{\theta}(x^{(i)}) - y^{(i)}) * x_2^{(i)})$

 K=k+1
 } update
 }
 return ($\theta_0^1, \theta_1^1, \dots$)

32 → 64
 → 128

All batches
 = 1 iter

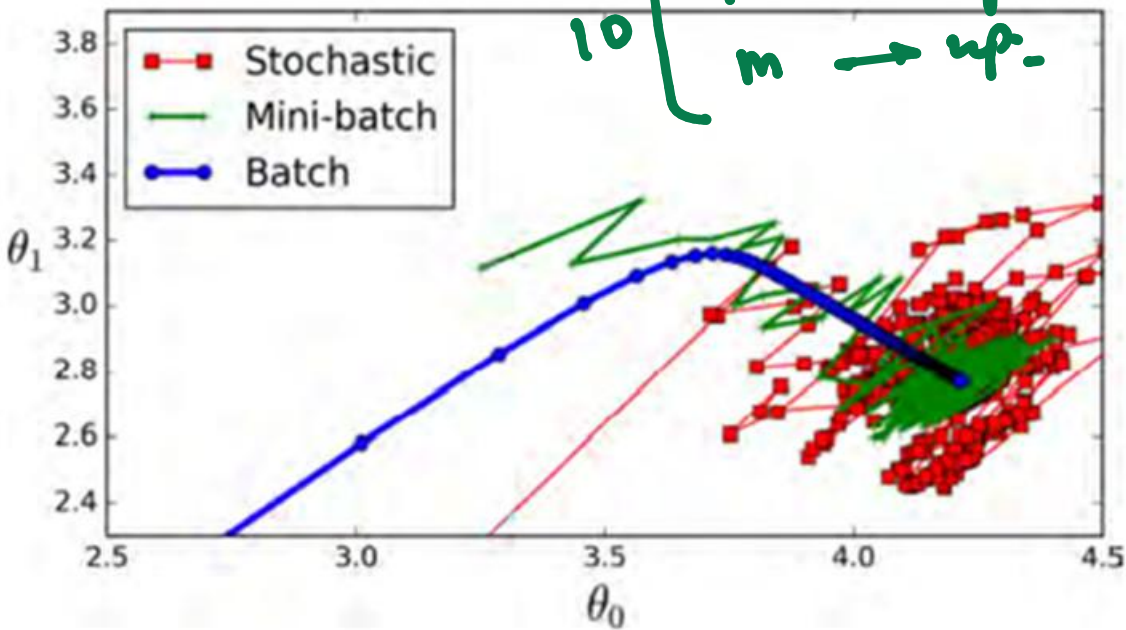
Gradient Descent: Variants - Effect on Convergence



Stochastic Gradient Descent

Batch = $\sum_{i=1}^d \rightarrow \text{update}$

✓ [Mini B = $\left[\begin{matrix} \sum_{i=1}^m \rightarrow \text{up.} \\ m \rightarrow \text{up.} \\ m \rightarrow \text{up.} \end{matrix} \right]$



1. Gradient Descent paths in parameter space

Evaluation Metrics

Evaluation of Linear Regression Model



Mileage (in kmpl)	Car Price (in cr)
9.8	10.48
9.12	1.75
9.5	6.95
10	2.51
....	

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2}$$

Unseen Data	
Mileage (in kmpl)	Car Price (in cr)
7.5	9.25
10	6.5
....	

$$R^2 = 1 - \frac{SS_{residual}}{SS_{Total}}$$

Co-eff. of determination

Model 1

$$\text{CarPrice} = 8.5 + 0.5 \text{ Mileage} - 1.5 \text{ Mileage}^2$$

Model 2

$$\text{CarPrice} = -5.5 + 1.5 \text{ Mileage}$$

goodness of fit

Evaluation of Linear Regression Model



R-squared

how much of var is explained by $\hat{y}$ y

Mileage (in kmpl)	Car Price (in cr)
9.8	10.48
9.12	1.75
9.5	6.95
10	2.51
....	

variation in 'y' that is explained by a regression model

explained variation = $\hat{y} - \bar{y}$ (actual, mean)

variation in 'y' that is not captured/explained by a regression model

unexplained variation = $y - \hat{y}$

total variation = $(y - \hat{y}) + (\hat{y} - \bar{y}) = (y - \bar{y})$

Car Price (in cr)
10.48
1.75
6.95
2.51
Mean Y

$R^2 = 1 - 0.02$
0.98

$R^2 = 1 - \frac{SS_{residual}}{SS_{Total}}$

Model 1

CarPrice = 8.5 + 0.5 Mileage - 1.5 Mileage²

$SS_{explained} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$

$SS_{residual} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$

$SS_{Total} = \sum_{i=1}^n (y_i - \bar{y})^2$

where :

Small
Var. low

- variation that is explained by a regression model
- measures the goodness of fit of a regression model

$SS_{explained}$ = explained variation sum of squares

$SS_{residual}$ = unexplained variation sum of squares

SS_{Total} = total variation sum of squares

$$\begin{array}{lcl}
 \theta \rightarrow h_{\theta}(x) \rightarrow \hat{y} & \text{---} & y \quad \checkmark \\
 \downarrow & & \\
 \theta' \rightarrow h_{\theta'}(x) \rightarrow \hat{y}' & \text{---} & y \quad \checkmark
 \end{array}
 \quad \begin{array}{l} \text{MSE, } R^2 \\ \text{MSE, } R^2 \end{array}$$

Linear Basis Models

32
Hy.1
 22x8
last []



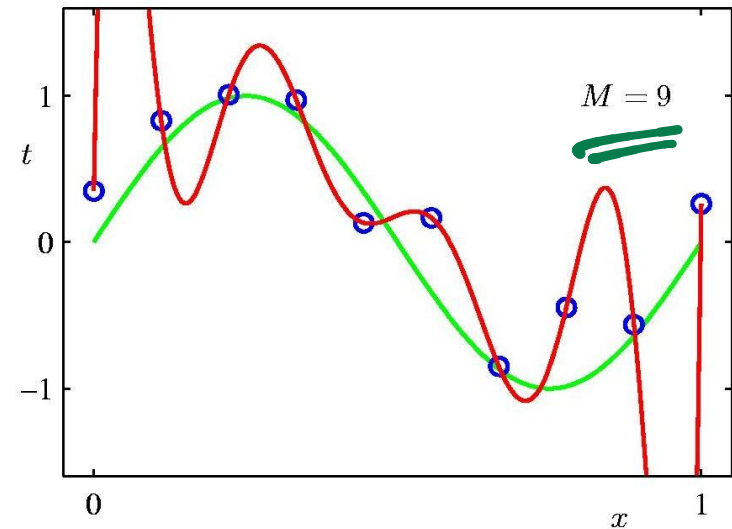
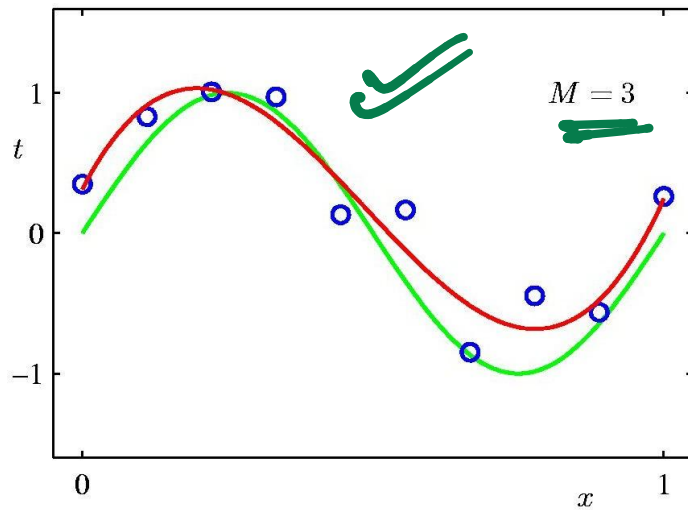
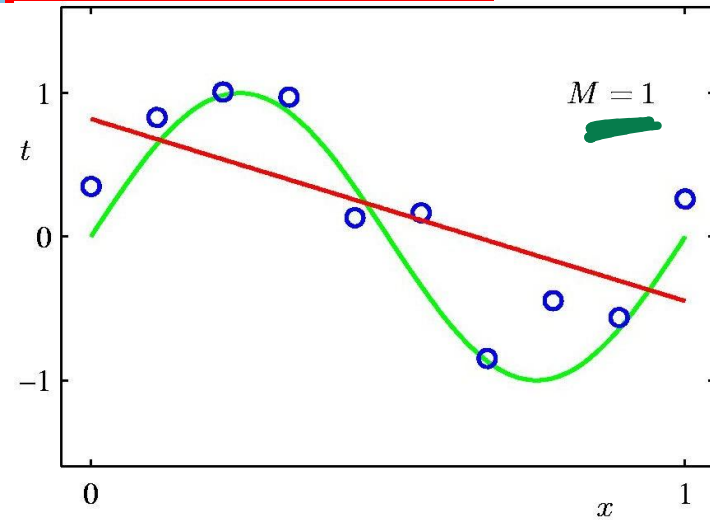
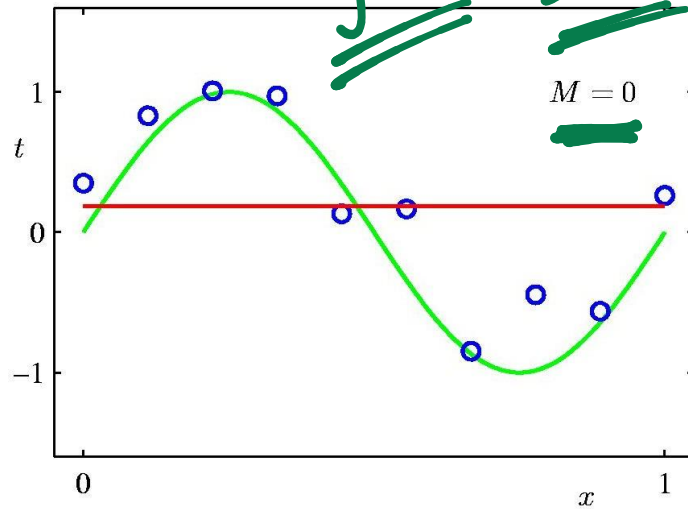
What if output is a non-linear function of input vector?

Polynomial Regression



$y = C$
 x^0
bias

x^1



Linear Basis Function Models



$\propto$

- The inputs X for linear regression can be:
 - Original quantitative inputs
 - Transformation of quantitative inputs
 - e.g. log, exp, square root, square, etc.
 - Polynomial transformation
 - example: $y = \beta_0 + \beta_1 \cdot x + \beta_2 \cdot x^2 + \beta_3 \cdot x^3$
 - Basis expansions
 - Dummy coding of categorical inputs
 - Interactions between variables
 - example: $x_3 = x_1 \cdot x_2$

Input X	Output Y
exp(2)	
exp(4)	
exp(6.3)	
exp(9.2)	

X No.of.Years of Experience (in Years)	<u>X^2</u>	Y Salary Of the Employee (in Lakhs)
1	1	2
2	4	3
3	9	4
4	16	5
5	25	6

X1 = Graduate	X2 = PostGraduate	X3 = Others	Y Salary Of the Employee
0	0	1	2
1	0	0	3
0	0	1	4
0	1	0	5
1	0	0	6

This allows use of linear regression techniques to fit non-linear datasets.

$0_0 + 0_1 x^3 \rightarrow x'$

$0_0 + 0_1 x'$

Compatibility
Leadership
Indiv

Assigned 1 = 18 May → Night 31 May

Webinar 1 - 19 May - 7:30 PM

$$\theta_0 + \theta_1 x_1 + \theta_2 x_2 \dots$$

higher
dep. /
low
exp

$$\theta_0 + \theta_1 x'_1 + \theta_2 x'_2 \dots$$

$\downarrow$ $\downarrow$

x_1 $\cos^2 x_2$

Lin Reg = Lin Basis
F_n

Linear Basis Function Models



Example: an M-th order polynomial function of one dimensional feature x :

$$y(x, \mathbf{w}) = w_0 + \sum_{j=1}^M w_j x^j$$

where x^j = j-th power of x

$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \dots + w_M x^M = \sum_{j=0}^M w_j x^j$$

$\phi_j(x)$ are known as *basis functions*. Typically, $\phi_0(x) = 1$, so that w_0 acts as a bias.

In the simplest case, we use linear basis functions : $\phi_d(x) = x_d$.

They are called linear models because this function is

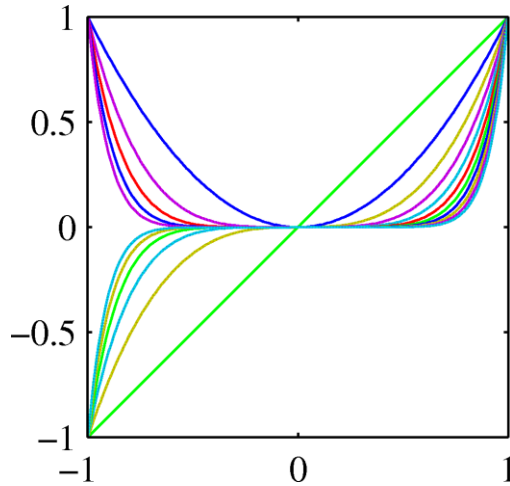
linear in $\mathbf{w}$.

X No.of.Years of Experience (in Years)	X^2	Y Salary Of the Employee (in Lakhs)
1	1	2
2	4	3
3	9	4
4	16	5
5	25	6

$$y(\mathbf{x}, \mathbf{w}) = \sum_{j=0}^{M-1} w_j \phi_j(\mathbf{x}) = \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x})$$

Tom 3 Sun
6 PM

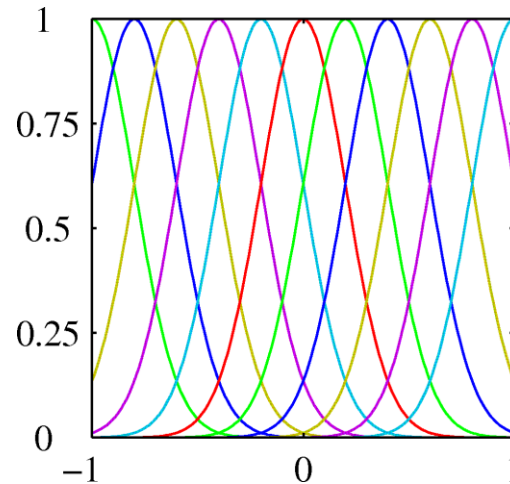
Linear Basis Function Models - Examples



Polynomial basis functions:

$$\phi_j(x) = x^j.$$

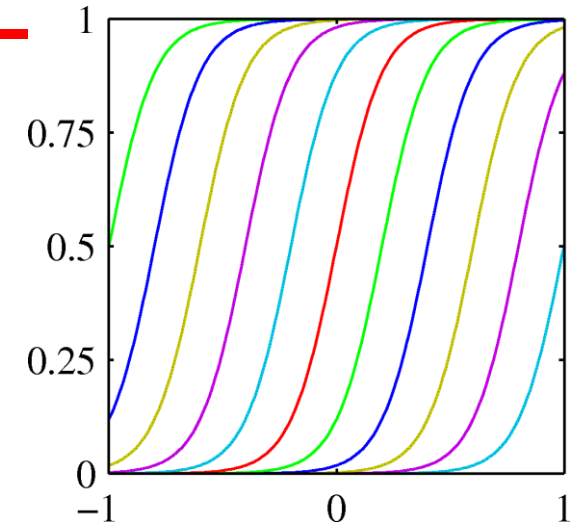
These are global; a small change in x affect all basis functions.



Gaussian basis functions:

$$\phi_j(x) = \exp \left\{ -\frac{(x - \mu_j)^2}{2s^2} \right\}$$

These are local; a small change in x only affect nearby basis functions. μ_j and s control location and scale (width).



Sigmoidal basis functions:

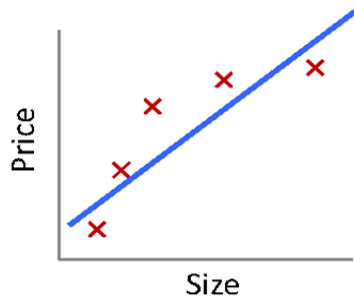
where $\phi_j(x) = \sigma \left(\frac{x - \mu_j}{s} \right)$

$$\sigma(a) = \frac{1}{1 + \exp(-a)}.$$

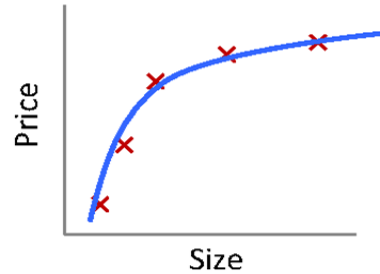
Also these are local; a small change in x only affect nearby basis functions. μ_j and s control location and scale (slope).

Notion of Bias - Variance

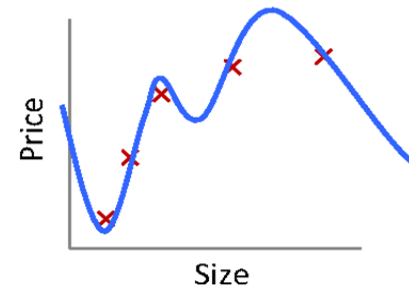
Quality of Fit



$\theta_0 + \theta_1 x$
Underfitting
(high bias)



$\theta_0 + \theta_1 x + \theta_2 x^2$
Correct fit



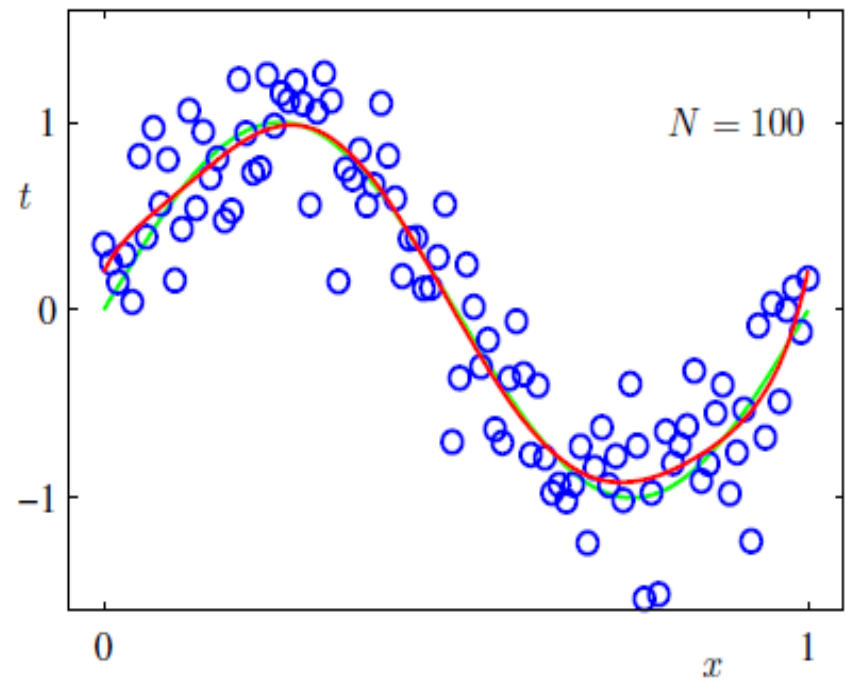
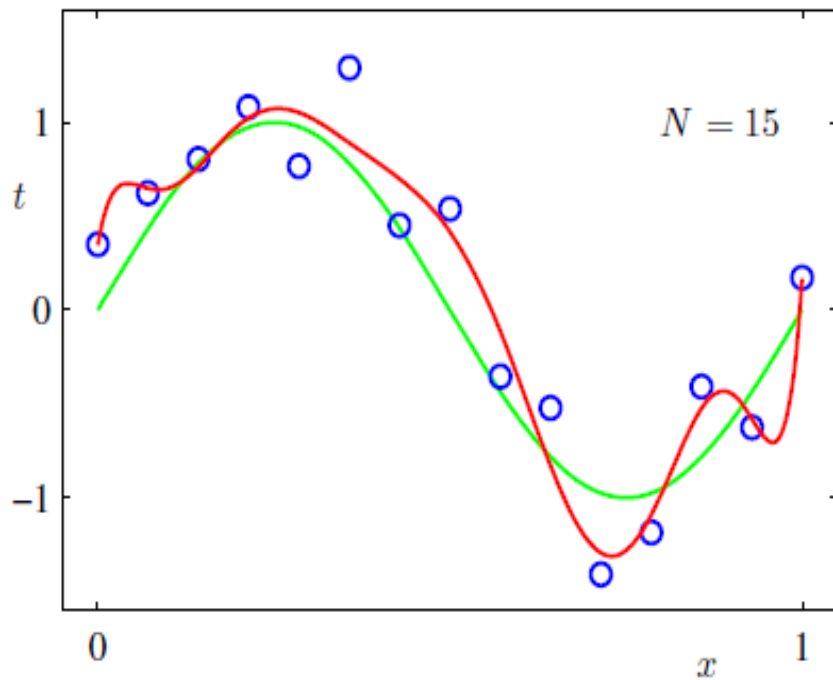
$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$
Overfitting
(high variance)

Overfitting:

- The learned hypothesis may fit the training set very well ($J(\theta) \approx 0$)
- ...but fails to generalize to new examples

Handling Overfitting – Way 1

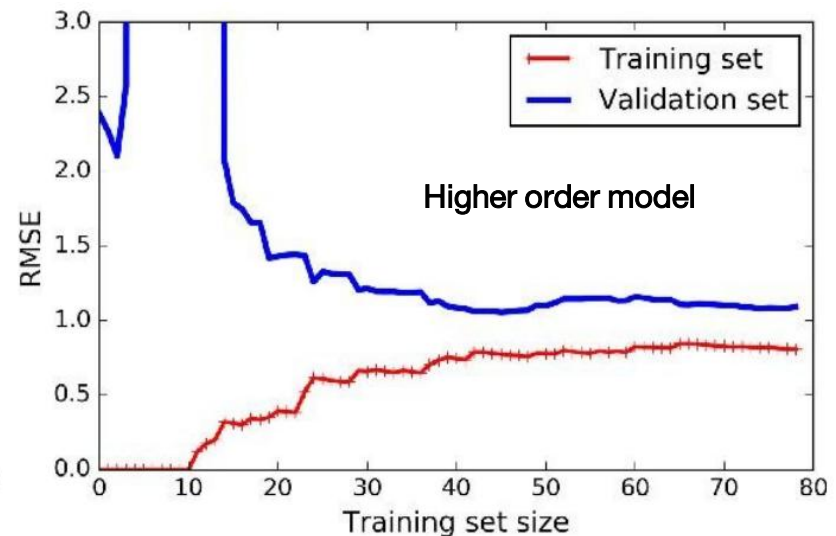
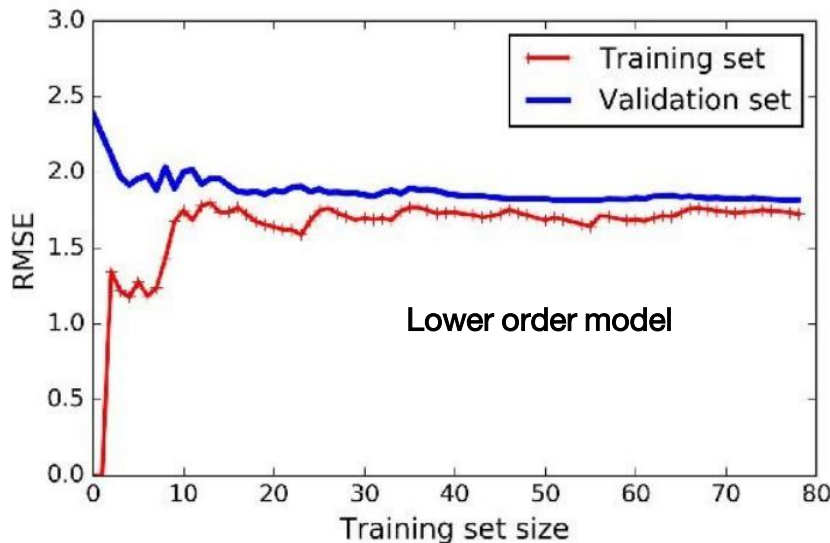
Increase in Size of the data set reduces the over-fitting problem



Effect of Training Size on Over fitting

Problem Type 4 : Interpretation of the Model Fit

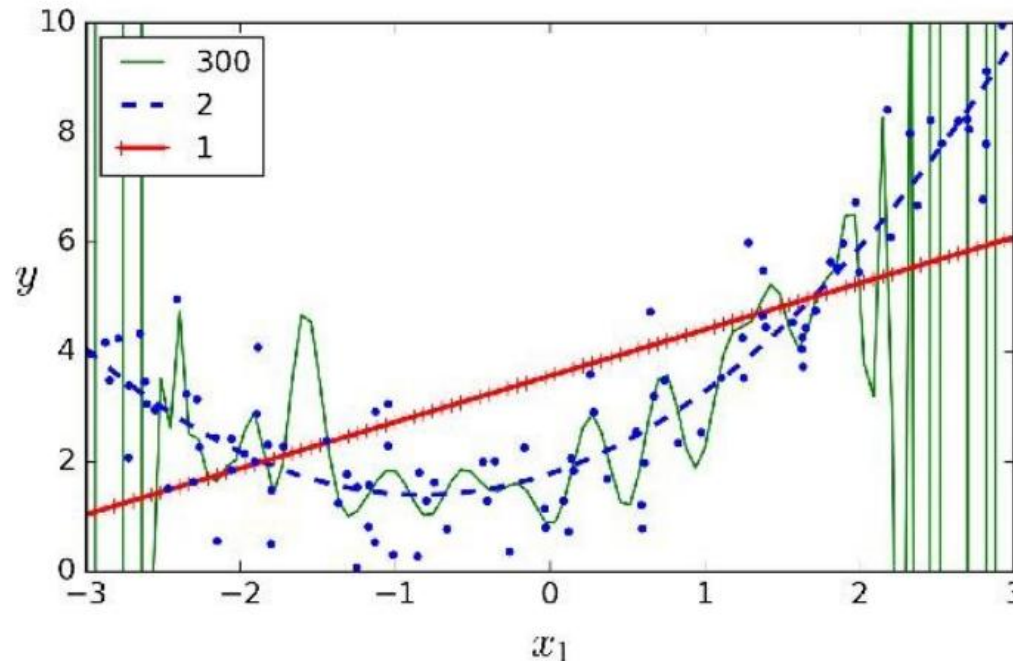
- Size of training dataset needs to be large to prevent overfitting when higher order model is used.



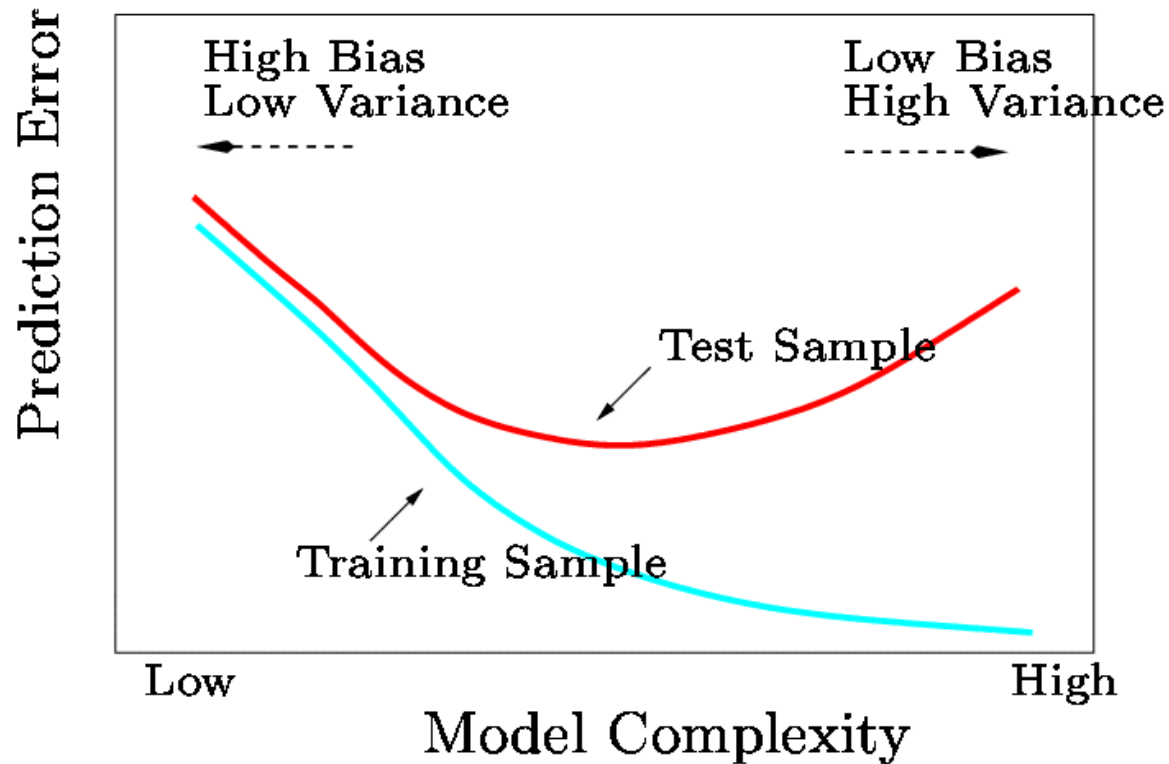
Polynomial Fitting can lead to Overfitting

Handling Overfitting – Way 2 – Reduce the complexity of the model

- Underlying target function is quadratic
- Linear model results in under fitting with large bias
- Polynomial of order 300 results in a large variance

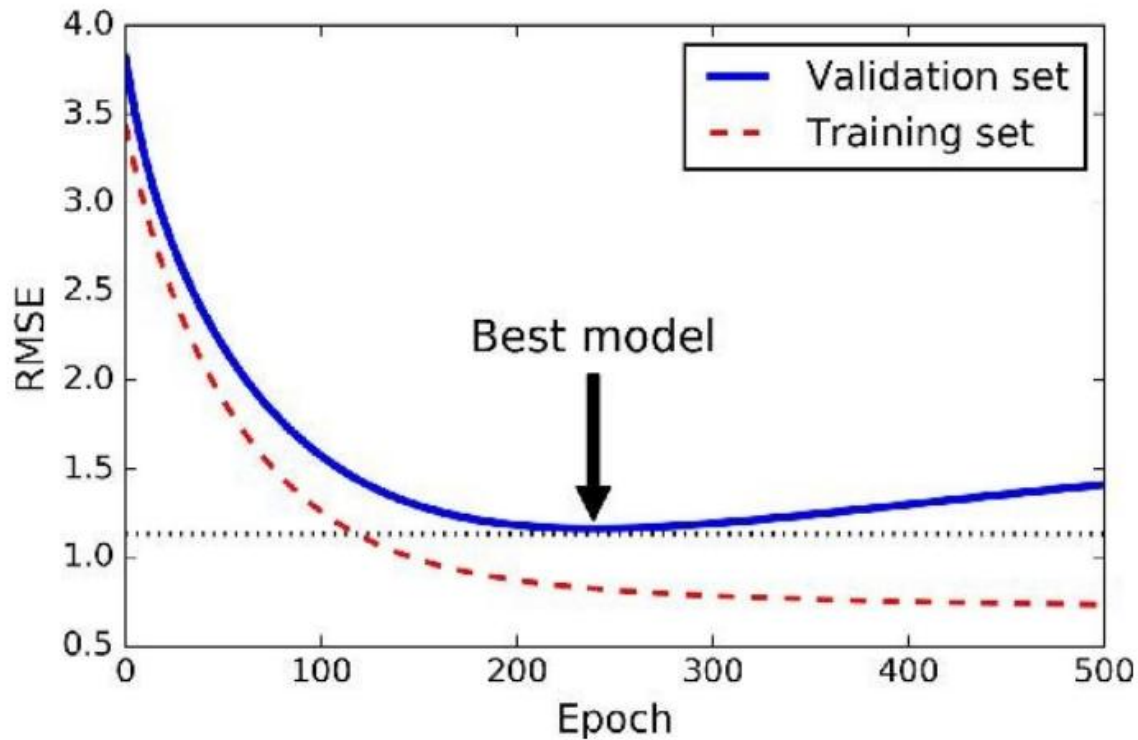


How to experiment on Model Complexity ?



Handling Overfitting – Way 3

Early Stop the Training



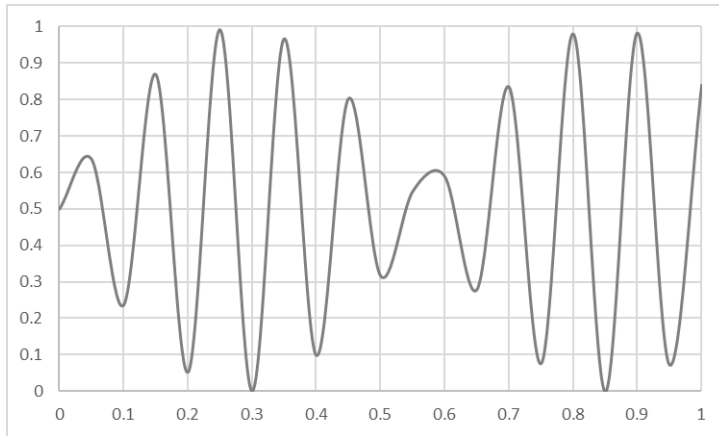
Regularization

(*This notion is common for both Linear Regression – Module 3 and Logistic Regression – Module 4)

Overfitting vs Underfitting

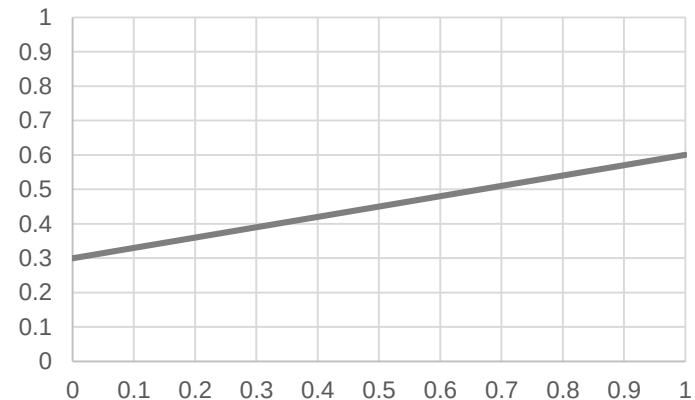
Overfitting

- Fitting the data too well
 - Features are noisy / uncorrelated to concept



Underfitting

- Learning too little of the true concept
 - Features don't capture concept
 - Too much bias in model



Regularization



- A method for automatically controlling the complexity of the learned hypothesis
- **Idea:** penalize for large values of θ_j
 - Can incorporate into the cost function
 - Works well when we have a lot of features, each that contributes a bit to predicting the label
- Can also address overfitting by eliminating features (either manually or via model selection)

Regularization



- Linear regression objective function

$$J(\boldsymbol{\theta}) = \underbrace{\frac{1}{2n} \sum_{i=1}^n \left(h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)} \right)^2}_{\text{model fit to data}} + \underbrace{\frac{\lambda}{2} \sum_{j=1}^d \theta_j^2}_{\text{regularization}}$$

- λ is the regularization parameter ($\lambda \geq 0$)
- No regularization on θ_0 !

Understanding Regularization



$$J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^n \left(h_{\boldsymbol{\theta}} \left(\mathbf{x}^{(i)} \right) - y^{(i)} \right)^2 + \frac{\lambda}{2} \sum_{j=1}^d \theta_j^2$$

- Note that $\sum_{j=1}^d \theta_j^2 = \|\boldsymbol{\theta}_{1:d}\|_2^2$
 - This is the magnitude of the feature coefficient vector!

- We can also think of this as:

$$\sum_{j=1}^d (\theta_j - 0)^2 = \|\boldsymbol{\theta}_{1:d} - \vec{\mathbf{0}}\|_2^2$$

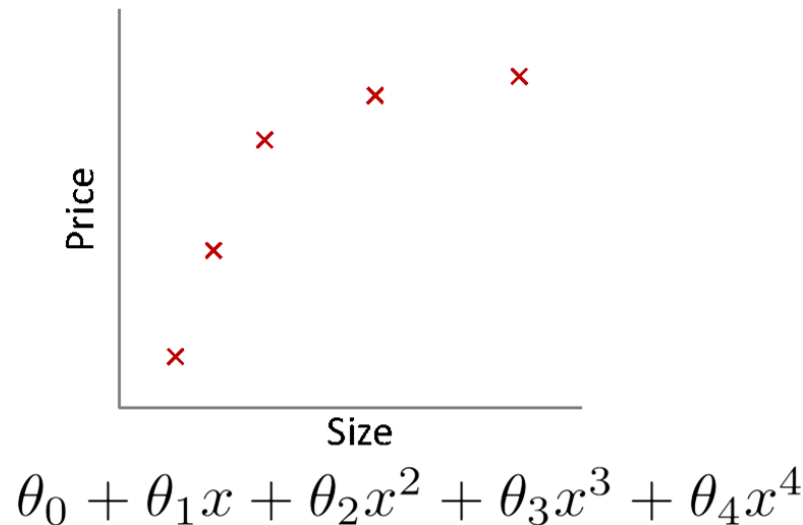
- L_2 regularization pulls coefficients toward 0

Understanding Regularization



$$J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^n \left(h_{\boldsymbol{\theta}} \left(\mathbf{x}^{(i)} \right) - y^{(i)} \right)^2 + \frac{\lambda}{2} \sum_{j=1}^d \theta_j^2$$

- What happens if we set λ to be huge (e.g., 10^{10})?



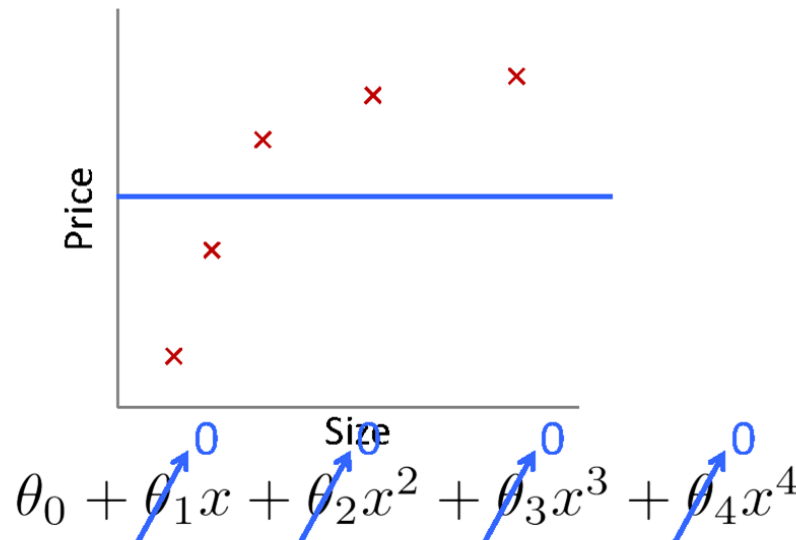
Based on example by Andrew Ng

Understanding Regularization



$$J(\theta) = \frac{1}{2n} \sum_{i=1}^n \left(h_{\theta} \left(x^{(i)} \right) - y^{(i)} \right)^2 + \frac{\lambda}{2} \sum_{j=1}^d \theta_j^2$$

- What happens if we set λ to be huge (e.g., 10^{10})?



Based on example by Andrew Ng

Regularization

Ridge Regression / Tikhonov regularization



- Cost Function
$$J(\theta) = \frac{1}{2n} \sum_{i=1}^n \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2 + \frac{\lambda}{2n} \sum_{j=1}^p \theta_j^2$$

- Fit by solving $\min J(\theta)$

- Gradient update:

$$\frac{\partial}{\partial \theta_0} J(\theta) \quad \theta_0 \leftarrow \theta_0 - \alpha \frac{1}{n} \sum_{i=1}^n \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)$$

$$\frac{\partial}{\partial \theta_j} J(\theta) \quad \theta_j \leftarrow \theta_j - \alpha \frac{1}{n} \sum_{i=1}^n \left(h_{\theta}(x^{(i)}) - y^{(i)} \right) x_j^{(i)} - \alpha \frac{\lambda}{n} \theta_j$$

Rewritten:
$$\theta_j \leftarrow \theta_j (1 - \alpha \lambda) - \alpha \cdot \frac{1}{n} \sum_{i=1}^n \left(h_{\theta}(x^{(i)}) - y^{(i)} \right) x_j^{(i)}$$

Regularization

Lasso Regression (Least Absolute Shrinkage and Selection Operator Regression)

- **Cost Function**

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^n \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2 + \lambda \sum_j |\theta_j|$$

- **Fit by solving $\min J(\theta)$**

- **Gradient update:**

$$\frac{\partial}{\partial \theta_0} J(\theta) \quad \theta_0 \leftarrow \theta_0 - \alpha \cdot \frac{1}{n} \sum_{i=1}^n \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)$$

$$\frac{\partial}{\partial \theta_j} J(\theta) \quad \theta_j \leftarrow \theta_j - \alpha \cdot \frac{1}{n} \sum_{i=1}^n \left(h_{\theta}(x^{(i)}) - y^{(i)} \right) x_j^{(i)} - \alpha \lambda \cdot \text{sign}(\theta_j)$$

$$\text{sign}(\theta_j) = \begin{cases} -1, & \theta_j < 0 \\ 0, & \theta_j = 0 \\ +1, & \theta_j > 0 \end{cases}$$

Regularization

Elastic Net



- **Cost Function**

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^n \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2 + r\lambda \sum_j |\theta_j| + \frac{(1-r)\lambda}{2} \sum_j \theta_j^2$$

Control the regularization using the **Mix Ratio “r”**:

$r = 0 \rightarrow$ equivalent to **Ridge Regression**;

$r = 1 \rightarrow$ equivalent to **Lasso Regression**

- **Gradient update:**

$$\frac{\partial}{\partial \theta_0} J(\theta) \quad \theta_0 \leftarrow \theta_0 - \alpha \cdot \frac{1}{n} \sum_{i=1}^n \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)$$

$$\frac{\partial}{\partial \theta_j} J(\theta) \quad \theta_j \leftarrow \theta_j - \alpha \cdot \frac{1}{n} \sum_{i=1}^n \left(h_{\theta}(x^{(i)}) - y^{(i)} \right) x_j^{(i)} - \alpha r \lambda \cdot \text{sign}(\theta_j) - \alpha(1-r)\lambda \theta_j$$

*from sklearn.linear_model import ElasticNet
ElasticNet(alpha=0.1, l1_ratio=0.5)*

How to choose the right Regularization?

Common Usage & Observation



- L1 regularization has the ability to set some coefficients to 0 exactly leading to a *sparse* model
- L1 regularization helps in feature selection by eliminating the features that are not important
- **L1 cannot be used efficiently in gradient-based approaches since it is not-differentiable unlike L2**
- L2 will in general lead to small magnitudes of weights but not exactly 0.
- Elastic net – Prefer in Highly correlated features

Numerical Exercise – For Student Practice

Fit a linear regression. Show only the first iteration of Gradient descent algorithm using learning rate of **0.02** for the following data, if the Relative Risk of Coronary Heart Disease is believed to be only linearly dependent on BMI as well as Diastolic Pressure. Assume the intercept of the regression model as **5** and the slope of independent variables as **-0.03 (negative)**.

Patient	Systolic Pressure mm Hg	Diastolic Pressure mm Hg	BMI	Waist Circumference Threshold cm	RR-CHD (Relative Risk of Coronary Heart Disease)
1	140	80	35	100	1.81
2	120	80	25	80	1.22
3	130	100	30	60	1.71

Apply a regularization on the same problem with regularization constant 5 and apply GD for 2 iterations & interpret the results. Try both ridge regression as well as lasso regression. Below equation is changed only for ridge regression. Students must apply appropriate GD update equation changes for this problem.

Steps :

1. Identification of the equations $y = w_0 + w_1X_1 + w_2X_2$
2. Cost function & derivative
 1. $W_0' = w_0 - 1/3 * \text{learning rate} * (\text{sum}(w_0 + w_1X_1 + w_2X_2 - y))$
 2. $W_1' = w_1 * (1 - \text{learning rate} * \text{regularization constant}) - 1/3 * \text{learning rate} * (\text{sum}(w_0 + w_1X_1 + w_2X_2 - y) * x_1)$
 3. $W_2' = w_2 * (1 - \text{learning rate} * \text{regularization constant}) - 1/3 * \text{learning rate} * (\text{sum}(w_0 + w_1X_1 + w_2X_2 - y) * x_2)$
3. Apply the equations



Python Lab Exercise – For Student Practice

Go to your virtual lab file under : Machine Learning ☐ LabCapsule 2 Linear _
Polynomial Regression ☐ 3 Polynomial Regression

Download ML_Lab 5 PolynomialRegression.ipynb

Try to change the degree to {2, 5, 7, 10} in the below function line :
`PolynomialFeatures(degree=3)`

Observe and interpret on the performance of the training data

Note: Training vs Test Data split is not coded in this implementation. Below are the suggestions to experiment further:

- Add few hundreds of data (Use synthetic data generation python libraries available : Refer here)
- Split the data into 80% train vs 20% test set
- Built the polynomial model using above four different degree on training set
- For each of the model apply in the test set
- Find the MSE for both training data and separately for test data
- Plot the four experiment results in a plot and interpret the notion of overfit vs underfit
- Suggest which degree is a best fit .

References

- T1 - Chapter 1 – Machine Learning, Tom Mitchell
- Chapter 1, 2 – Introduction to Machine Learning, 2nd edition, Ethem Alpaydin
- R1 – Chapter #1, # 3,#4 (Christopher M. Bhisop, Pattern Recognition & Machine Learning) & Refresh your MFDS course basics

Thank you !